Resource-backed Loans and Sovereign Debt Default in Developing Countries: Do institutions matter?

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Abstract

This paper uses the entropy balancing method to test whether resource-backed loans have a causal effect on sovereign debt default in a large sample of 134 developing countries over the period 1990-2020. Through a battery of robustness checks, the estimation results indicate that resource-backed loans have a positive and statistically significant effect on sovereign debt default. When we disaggregate resource-backed loans to run the regressions, the results show that all three types of resource-backed loans have a positive and statistically significant effect on sovereign debt default, with a greater impact of cocoa, tobacco-backed loans and oil-backed loans than that of mineral-backed loans. Another result put forward in this paper shows that the amplifying effect of resource-backed loans is mediated by low institutional quality and borrowing costs. Finally, our results show strong heterogeneity when certain structural, political and institutional factors are controlled. The main conclusions of the study suggest good transparency in resource-backed loans arrangements, and call for vigilance on the part of policy-makers in borrowing countries when it comes to the accumulation of resource-based loans.

Keywords: • Resource-backed loans • Resource rents • Sovereign debt • Entropy balancing.

JEL Codes: O13, H81, C12, H63.

1 Introduction

Achieving the Sustainable Development Goals requires considerable investment, particularly in infrastructure, human capital and resilience to climate change. Yet, in developing countries, governments often have limited means to mobilize public revenues or private investment (Kose et al., 2021). Moreover, many of them are unable to better mobilize resources on the financial market, as their ability to access capital markets has been limited over the past two decades (Mihalyi et al., 2022). From then on, resource-rich countries will use the potential product flows from their natural resource windfalls as collateral or in exchange for access to funds (Mihalyi and Scurfield, 2021; Wang et al., 2023; Manzano and Rigobon, 2001; Mihalyi et al., 2020), a new financing model known as «resource-backed loans» developed during the natural resource booms of the 2000s (Mihalyi et al., 2020; Wang et al., 2023). By definition, resource-backed loans are financing's provided to a government or public enterprise, with repayment either directly in the form of natural resources (in kind), such as Oil or minerals, or from revenues generated by those resources, or repayment is secured by future revenues related to the natural resources, or the reserves of the natural resources serve as collateral for the loan (Mihalyi et al., 2020, 2022).

Geographically speaking, African and Latin American countries are the main recipients of resource-backed loans (Coulibaly, 2023; Mihalyi et al., 2020, 2022)[1]. According to Mihalyi et al. (2020), between 2004 and 2018, 52 resource-backed loans were signed in developing countries. During this period, 30 of the 52 loans were signed by African countries, while the remaining 22 were signed by Latin American countries. Technically, the composition of these 52 resource-backed loans identified on natural resources indicated that 43 of these loans were backed by oil, 6 by minerals, 2 by cocoa and 1 by tobacco. The total amount of resource-backed loans signed over the period 2004-2018 was valued at \$164 billion, of which \$98 billion went to Latin American countries and \$66 billion to sub-Saharan African countries (Mihalyi et al., 2020, 2022).

In addition, the fuller composition of loans indicates that Guinea is Africa's largest recipient of resource-backed loans in 2017 (Mihalyi et al., 2020). In 2017, the country signed a \$20 billion resource-backed loan (nearly 200% of the country's GDP) with a consortium of Chinese institutions (Mihalyi et al., 2020). The second largest recipient of resource-backed loans in Africa is the Republic of Congo, which signed a \$1.6 billion resource-backed loan in 2006, equivalent to 21% of the country's GDP that year. Close behind was the Democratic Republic of Congo's Sicomines infrastructure agreement signed with China in 2008. The loan amounted to 3 billion USD, equivalent to 16% of the country's GDP (Mihalyi et al., 2020, 2022). While Venezuela is the biggest beneficiary among

[1] These countries are : Angola, Brazil, Chad, Democratic Republic of Congo, Ecuador, Ghana, Guinea, Niger, Republic of Congo, Sao Tome and Principe, Sudan, South Sudan, Venezuela and Zimbabwe (Mihalyi et al., 2020; Coulibaly et al., 2022; Coulibaly, 2023).

Latin American countries. Between 2004 and 2018, the country signed around $59 billion in resource-backed loans. Next come Brazil and Ecuador with $25 billion and $13.8 billion respectively (Mihalyi et al., 2020). Provided it is well managed, transparent and used as part of a credible growth policy, debt can be a lever. If this were not the case, we would be witnessing what Gonzalez-Redin et al. (2018) calls a debt-based economy, doomed to disaster.

In indeed, while both advanced and developing countries continue to commit to increasing public investment to achieve sustainable development goals through the exploitation of natural resources, policymakers have not emphasized the importance of assessing the likely macroeconomic effects resulting from public investment flows in countries with a rich subsoil in natural resources (Wang et al., 2023). In recent years, however, debt-related vulnerabilities have been on the rise in emerging markets and low-income countries (Kose et al., 2021). During the end of 2019, the outstanding external debt of low- and middle-income countries totaled $8.1 trillion, with a third of this amount held by private creditors (Kose et al., 2021). Many African countries such as DR Congo, Egypt, Ghana, Mozambique, Namibia, South Africa, Tanzania, Zambia, Zimbabwe etc., although endowed with a wealth of natural resources, are generally choked by a plethora of foreign debt (Wang et al., 2023). Moreover, according to the recent report on list of LIC DSAs for PRGT Eligible Countries, by the IMF, ten (10) countries are currently in debt distress and nine of these are from Africa (ECA, 2022)[2]. Given the current global situation, the issue of debt restructuring is necessary to reduce the debt burden to sustainable levels and limit the negative impact of higher debt servicing on an already fragile socio-economic environment in heavily indebted countries, particularly those in Africa (ECA, 2022).

Furthermore, sovereign debt restructuring precedes sovereign debt default, which is a function of debt levels and fiscal discipline. Basically, sovereign debt default concerns countries that have experienced one or more sovereign defaults (on external or domestic debt, public or state-guaranteed, or both)[3]. Somewhat abruptly and unexpectedly, several developing countries are currently undergoing preventive debt restructuring, which would help avert a debt crisis with disastrous socio-economic repercussions and provide additional fiscal room for maneuver as countries recover from the coronavirus pandemic (COVID-19) and weather the food and energy price shocks resulting from the conflict between the Russian Federation and Ukraine (ECA, 2022). Indeed, many resource-rich countries are facing a situation that resembles what Prebisch (1962); Singer (1950); Destek et al. (2023); Badeeb et al. (2023); Liang et al. (2023); James and Aadland (2011); Papyrakis and Gerlagh (2007); Sachs and Warner (1999); etc., have called the resource curse. So, it makes sense to analyze in depth the causes and even the consequences of a policy of indebtedness linked to natural resources. Authors such as Sawadogo (2020) believes that

[2]They include Chad, the Republic of Congo, Malawi, Mozambique, Soa Tome and Principe, Somalia, Sudan, Zambia and Zimbabwe (see, IMF, 2021, 2022).

[3]These defaults may occur five or fifty years apart; they may be total defaults (or repudiation) or partial defaults through rescheduling (Reinhart and Rogoff, 2011).

the cause of these countries' poor macroeconomic indicators is fiscal indiscipline, and that the adoption of fiscal rules could help them to deter fiscal profligacy by strengthening the credibility of their fiscal policy and reducing borrowing costs. Others believe it's the over-indebtedness of resource-rich countries resulting from resource-backed loans (Mihalyi et al., 2020, 2022; Wang et al., 2023; Mihalyi and Scurfield, 2021). In fact, more loans can be obtained since the natural resources of these countries serve as collateral. As a result, investors are discouraged from granting capital, leading these countries into the debt trap of high levels of indebtedness resulting from unfettered borrowing by resource-rich regimes (Wang et al., 2023). According to Mihalyi et al. (2020), resource-backed loans could exacerbate public debt sustainability and lead countries into debt distress. The authors explain that ten (10) of the fourteen (14) countries that have signed up to resource-backed loans have been faced with a plethora of external debt. From then, it makes sense to analyze the link between resource-backed loans and sovereign debt default in developing countries.

Given that resource-backed loans are generally underwritten in US dollars, the collapse in natural resource prices could increase the nominal amount of these loans (Mihalyi et al., 2022) and lead countries into a situation of over-indebtedness (Mihalyi et al., 2020). As a result, these countries experience what is known as debt overhang and hence sovereign debt default — a situation where huge debt levels peak when the regimes of these countries are unable to generate enough revenue to settle debts and finance forward-looking expenditures (Ampofo et al., 2021). In the same vein, repayment of resource-backed loans could lead governments either to undertake oil resource exploitation, or to accelerate the pace of mineral and hydrocarbon production to meet the repayment schedule, forcing them to focus on the mining sector and resource rents. Whereas natural resource revenues are generally associated with exchange rate appreciation, leading to a loss of competitiveness in non-resource market sectors, leading to Dutch systems (Van der Ploeg and Venables, 2013). This would pose problems for fiscal and macroeconomic policy management (Arezki and Nabli, 2012; Anyanwu and Erhijakpor, 2014; Ross, 2015).

In addition, resource-backed loans can jeopardize the public debt situation (Horn et al., 2021) and may prevent the country from raising financial resources to fund sustainable development policies. Fundamentally, unsustainable public debt reduces fiscal space and weakens the government's ability to undertake public investment, fight inequality and eradicate poverty. Resource-backed loans can worsen a country's debt sustainability (Horn et al., 2021), which is likely to adversely affect economic activity (Reinhart and Rogoff, 2010; Panizza and Presbitero, 2014), and thus lead to a situation of debt distress. What's more, due to the lack of transparency surrounding resource-backed loans (Mihalyi et al., 2020; Horn et al., 2021; Rivetti, 2021), countries are exposed to real — dangers of sovereign debt default as resource-backed loans may not be reflected in public debt statistics (Mihalyi et al., 2020; Horn et al., 2021; Rivetti, 2021). In the same vein, resource-backed loans can lead to a deterioration in institutional quality, affect democratization and precipitate poor policy choices that lead to inefficient resource transfers.

However, it is possible that the financial resources of resource-backed loans will be used to finance the government budget (Mihalyi et al., 2020), thus avoiding a situation of sovereign debt default. Indeed, resource-backed loans are generally invested in public infrastructure projects (roads, schools, hospitals) that can stimulate domestic aggregate demand and generate revenue for sovereign debt repayment over the medium and long term (Coulibaly et al., 2022). In addition, resource-backed loans can facilitate access to financial markets to mobilize founds for development financing, and boost economic activity through massive productive investment. Resource-backed loans can be renegotiated in the event of payment difficulties (Mihalyi et al., 2020, 2022), which could help avoid sovereign debt default. The repayment structure of some RBLs allows the government to pay less in monetary terms when commodity prices (or production levels, or profits) are low, and to repay the loan more quickly when conditions are favorable (Mihalyi et al., 2020), this avoids over-indebtedness and limits the probability of sovereign debt default.

Resource-backed loan has received considerable attention in the existing literature in recent years. Early studies, essentially non-empirical, present resource-backed loans as an honored financing tool that favors access to capital markets (Manzano and Rigobon, 2001; Mihalyi et al., 2020; Mihalyi and Scurfield, 2021). In addition, other studies explore the effect of resource-backed loans on forest loss cover (Coulibaly, 2023), medium- and long-term debt reduction (Coulibaly et al., 2022), pitfalls and opportunities (Mihalyi et al., 2020, 2022). Although macroeconomic studies remain scarce, some studies identify resource-backed loans as a mechanism both financing and over-indebtedness of the economy (Mihalyi et al., 2022), which negatively affects institutional quality due to its opacity and lack of transparency (Rivetti, 2021; Horn et al., 2021), and jeopardizing the sovereign debt situation (Horn et al., 2021; Mihalyi et al., 2020; Mihalyi and Scurfield, 2021). Despite the evolving literature on resource-backed loans and its effects on the economy, to our knowledge, little has been said about fiscal procyclicality empirically. This paper aims to fill this gap by analyzing whether resource-linked debt such as resource-backed loans can help developing countries to finance their development efficiently without raising the risk of sovereign debt default.

On the basis of the literature, we hypothesize that resource-backed loans can have an impact on sovereign debt default through two main channels, namely the cost of sovereign debt and corruption. The cost of debt channel is justified by what Wang et al. (2023) call the public debt trap. In fact, in times of soaring natural resource prices, creditors develop an interest in granting loans since the lending capacity of booming countries could be improved. As a result, more loans can be obtained as these nations' natural resources serve as collateral. This would reduce the cost of public debt due to the guarantee of repayment. The second channel is the weak institutional quality captured by corruption. Corruption reduces the revenue available to repay debt, while RBL increases the amount of debt to be repaid through untraced loans (such as resource-backed loans, which do not appear in the budget statistics of signatory countries). Ultimately, corruption and unpaid loans reinforce each other, leading to debt distress. Thus, the opacity of RBLs allows corruption to flourish, and corruption scares off traditional credit markets and encourages recourse to RBLs. This builds up the stock of debt to a level where the country defaults on its sovereign debt. Moreover, this channel of corruption is justified by RBL's terms and conditions, which raise concerns about governance, corruption, and transparency in the context of agreed transactions (Mihalyi et al., 2020, 2022).

To identify the causal effect of resource-backed loans, we rely on entropy balancing, an impact analysis method, developed by Hainmueller (2012). Using a sample of 134 developing countries over the period 1990–2020, we show that resource-backed loans adoption increases the sovereign debt default in resource-backed loans countries, relative to non resource-backed loans countries. This result is robust to several robustness tests, including a placebo test, changing the definitions of sovereign debt default, adding additional control variables, changing the sample design, and using alternative estimation methods such as propensity score matching, propensity score weighting, inverse probability weighting, coarsened exact matching, higher moments (variance) adjustment, panel fixed effects, and system GMM. To explain this result, we shed light on different transmission channels. We show that the negative impact of resource-backed loans may stem from their ability to increase the cost of debt in financial markets and weaken countries' institutional quality. Then, heterogeneity tests show that the impact of resource-backed loans may depend on the duration, the type of resource-backed loan and certain structural factors, in particular, the volatility of government spending, inflation targeting, foreign direct investment, remittance inflows, resource depletion, democracy, rule of law, and political stability.

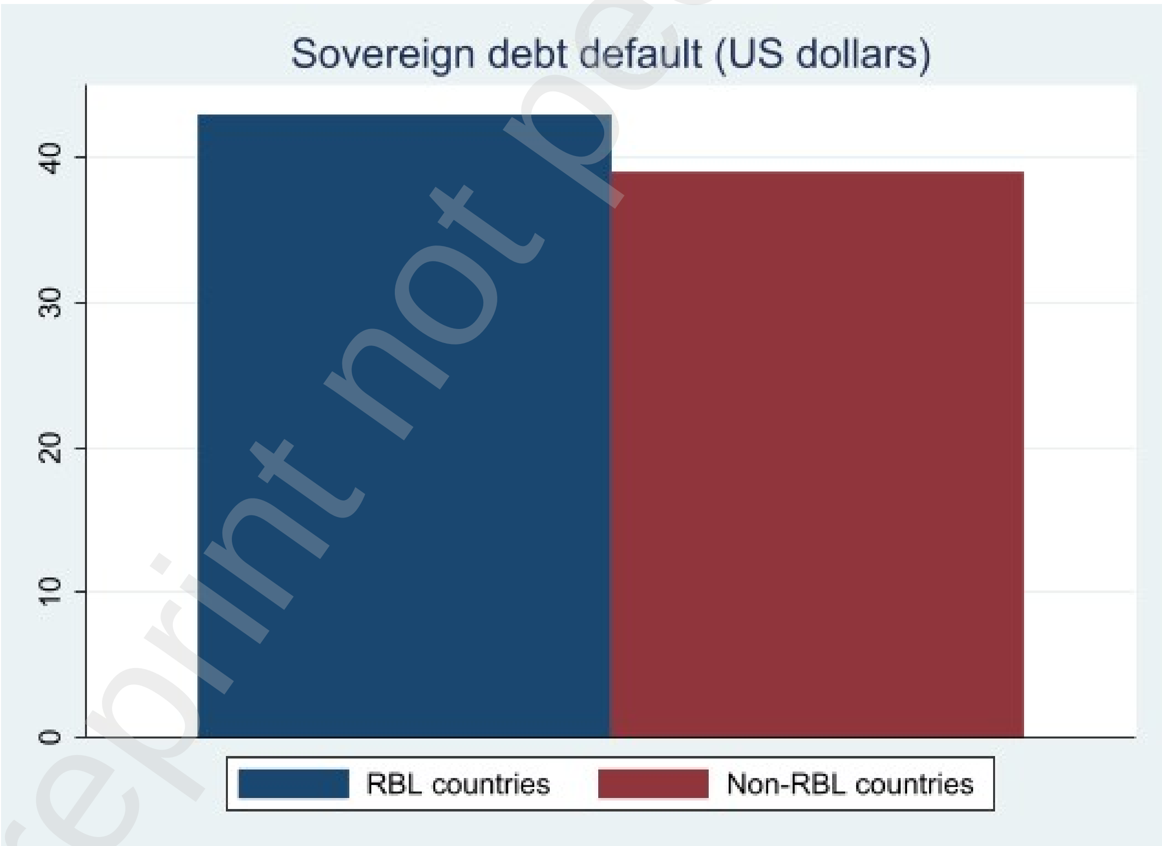
The rest of the paper is organized as follows. Section 2 reports some stylized facts. Then, Section 3 describes the empirical methodology, while Section 4 presents the data and descriptive statistics. The main results are presented in Section 5, while in Section 6 -7 we examine the sensitivity of these results and analyses the possible transmission channels. Section 8 explores the heterogeneity tests. Finally, Section 9 concludes the study and presents the main policy recommendations from the results.

2 Stylized facts

This section presents some stylized facts that characterize the relationship between resource-backed loans (RBL) and the average evolution of the sovereign debt default over the period 1990-2020. The statistics cover 134 developing countries, with 14 RBLs and 120 non-RBLs countries. Figure 1 below shows, on average, a higher sovereign debt default in RBLs countries compared to non-RBLs. Figure 2 presents the average evolution of the sovereign debt default for RBLs and non-RBLs, before and after adopting RBL. We follow the methodology used by Mishkin and Schmidt-Hebbel (2007) and Minea and Tapsoba (2014) to construct sovereign debt default before and after RBLs adoption for non-RBLs. Figure 2 shows an increase in the sovereign debt default in both groups of countries after RBL adoption. However, this increase was substantial in RBLs compared to non-RBLs. Indeed, in RBLs, the sovereign debt default increases from an average of 3% before RBL adoption to 6% after RBL adoption, while this rate increases from 3.7 to 3.9% among non-RBLs. Thus, the evolution of the sovereign debt default after the adoption of RBL is about three times greater in RBLs compared to non-RBLs (+3% versus +0.2%).

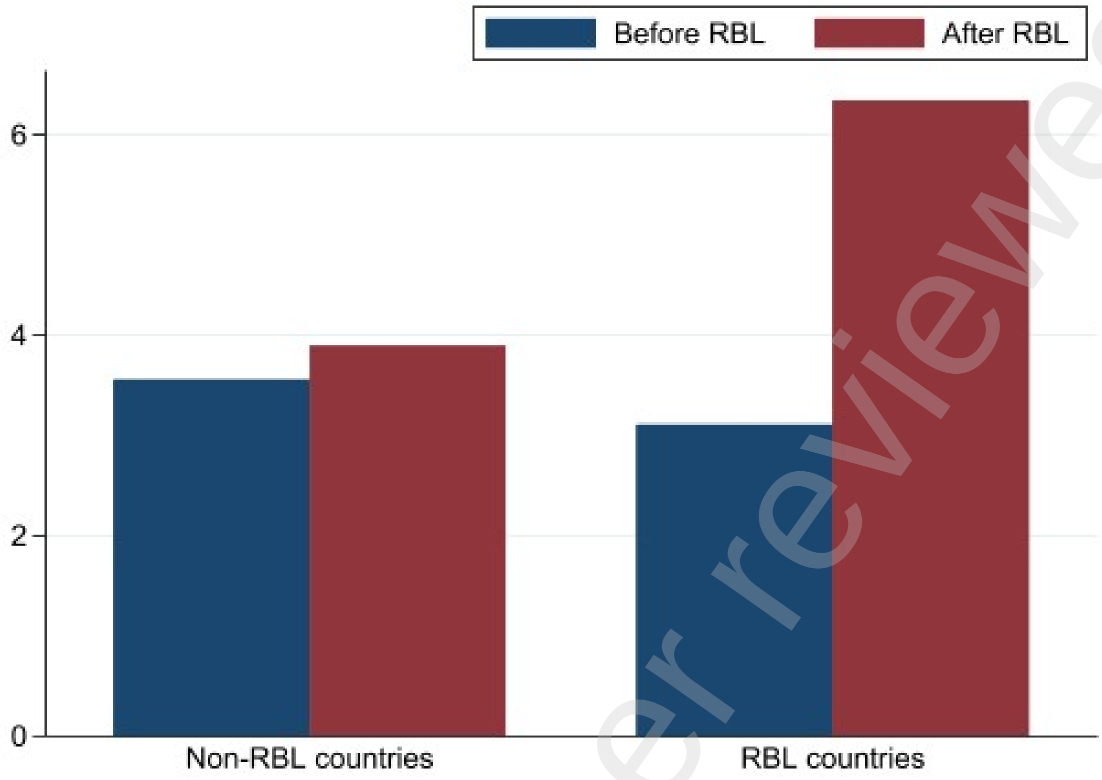
Thus, Figure 2 highlights a striking fact. Although both groups experienced increased sovereign debt default after adopting RBL, the gap between RBLs and non-RBLs countries widened, with a significant difference of around 2 percentage points. This graphical evidence shows that the adoption of resource-backed loans is likely to go hand in hand with sovereign debt default. This allows us to establish a correlation between our outcome variable and the treatment variable, but beware this does not provide any causal relationship. For this reason, an econometric impact analysis is envisaged in this study to estimate causal effects between the treatment variable and the outcome variable.

Figure 1: Average sovereign debt default in RBL and non-RBL



Source: Author's elaboration using data from Beers and Mavalwalla (2017).

Figure 2: **Average sovereign debt default before and after RBL adoption.**



Source: Author's elaboration using data from Beers and Mavalwalla (2017).

3 Methodology

Our aim is to examine the causal effect of resource-backed loans on sovereign debt default. Indeed, identifying and tracing down precisely any genuine effects induced by resource-backed loans on the macro-variables of interest is challenging, given the potential selection problem that arises from the fact that signing resource-backed loans may be correlated with factors that also affect the country's budget policy. In fact, the reasons why developing countries sign up to resource-backed loans may be linked to macroeconomic conditions, economic performance, debt levels, fiscal overhang, limited access to financial markets, etc. Although these factors may affect sovereign debt default, they make adoption endogenous. While these factors may affect sovereign debt default, they make adoption endogenous. If resource-backed loans were exogenous, using a simple fixed-effects model would be quite sufficient to identify its effects. We cannot go into this problem of potential endogeneity of resource-backed loans in view of the interdependence's between them and macroeconomic results, otherwise it would lead to a bias in the estimates.

We address this endogeneity using a matching approach, which consists in estimating the average treatment effect on the treated (ATT), defined as follows :

$$ATT{=}E[Y_{i(1)}|T_i{=}1]\text{-}E[Y_{i0}|T_i{=}1] \tag{1}$$

where $Y_{i(.)}$ is the outcome variable measuring the sovereign debt default. T_i indicates whether the observation unit is subject to resource-backed loans adoption ($T_i = 1$) or not ($T_i = 0$). $E[(Y_{i(1)}|T_i{=}1]$ is the sovereign debt default level during the resource-backed loans adoption period, $E[(Y_{i0}|T_i{=}1)]$ is the counterfactual result for countries that had adopted resource-backed loans, i.e. the sovereign debt default in countries that had adopted resource-backed loans if they had not.

Equation 1 implies that comparing sovereign debt default observed in countries without resource-backed loans with sovereign debt default that would be observed in the same countries if resource-backed loans were signed would yield an unbiased estimate of ATT. The main difficulty, however, is that the second term on the right-hand side of this equation is unobservable. We cannot observe the sovereign debt default of a country without resource-backed loans if it had signed resource-backed loans. In this case, given a random selection of non-resource-backed loans countries, we can simply compare the sample average of the non-resource-backed loans countries and the resource-backed loans countries to avoid this problem. However, as previously explained, the choice to sign resource-backed loans may be driven by certain observable factors (economic performance, macroeconomic conditions, economic performance, debt levels, fiscal overhang, limited access to financial markets, etc.) that also determine sovereign debt default. This can lead to self-selection.

In addition, the comparing average value of sovereign debt default between the two samples may lead to a problem of ***"selection on observables"*** and bias the linear regression method (Balima and Sy, 2021). So, estimate of the ATT under unconfoundedness (or conditional independence), meaning that systematic differences in outcomes between treated and comparison individuals with the same covariate values are attributable to treatment, we can replace the unobservable term $E[(Y_{i0}|T_i{=}1)]$ in Equation 1 with the observable term $E[Y_{i0}|T_i{=}1, X = x]$. Considering these various elements and proofs, we can rewrite Equation 2 as follows:

$$ATT{=}E[Y_{i(1)}|T_i{=}1, X = x]\text{-}E[Y_{i0}|T_i{=}1, X = x] \tag{2}$$

where $X = x$ is a vector of observable covariates that may affect both the decision to sign resource-backed loans and the sovereign debt default, $E[Y_{i(1)}|T_i{=}1, X = x]$ is the sovereign debt default of resource backed loans units, and $E[Y_{i0}|T_i{=}1, X = x]$ is the expected sovereign debt default for the synthetic control units. Following a handful of recent impact assessment studies, I use entropy balancing developed by Hainmueller (2012) and implemented by Neuenkirch and Neumeier (2016) and Balima and Sy (2021). This method consists of two principal steps: First, the weights assigned to the control units (in our case, the non-resource-backed loans countries) are calculated in the entropy balancing. In the second step, the weights obtained in the first step are used in a regression analysis with the treatment variable (resource-backed loans countries) as the explanatory variable. We then weight the countries with and without resource-backed loans based on observable characteristics. Thus, the average difference in sovereign debt default between countries with resource-backed loans and the ***"closest"*** non-resource-backed loans countries should be explained by the signing of resource-backed loans.

Unlike other methods for estimating effects and treatment effects, such as propensity score matching, entropy balancing has several advantages because of its ability to combine both matching and regression analysis (see, for instance, Hainmueller (2012); Balima and Sy (2021); Sawadogo (2020); and Apeti (2023)). In fact, this method outperforms the classical approach based on regression and matching based on propensity score methods because it is non-parametric — which allows to avoid the problem of mis-specification of the functional form of the model that could bias the results. Moreover, it is a method that also eliminates multicollinearity problems, since the mechanism of re-weighting makes the treatment variable orthogonal to the covariates (Hainmueller, 2012).

This method has been used by several authors in their studies because of its advantages. For instance, Balima and Sy (2021) to examine the role of IMF-supported programs in reducing the likelihood of subsequent sovereign defaults in borrowing countries. Apeti (2023) to assess the effect of mobile money adoption on household consumption volatility. Apeti and Edoh (2023) uses it to assess the impact of mobile money on tax revenue mobilization. Jacolin et al. (2021) examine the impact of mobile financial services in particular, mobile money, mobile credit, and savings on the informal sector.

These studies explain the advantages of entropy balancing in impact studies over other methods, which reinforces our trust that the method is robust. The first advantage of the method is that unlike propensity score matching or difference-in-difference estimators, entropy balancing is a non-parametric approach, thus requiring no specification of the functional form of the empirical model or of the treatment assignment procedure, which may avoid specification errors. Second, the weight system orthogonality the covariates with respect to the treatment, which limits multicollinearity issues. Third, the method ensures a suitable balance of pre-treatment characteristics between the treatment and control groups, even in the presence of a small sample or a limited number of untreated units. Another advantage is that entropy balancing uses more flexible reweighting schemes compared to conventional balancing, where control units are either discarded or balanced. Here, units are reweighted with the goal of achieving a balance between processed and unprocessed units, keeping the weights as close as possible to the base weights to avoid information loss. Its most attractive feature is to allow a high degree of covariate balance between resource-backed loans countries and non resource-backed loans countries—even in small samples—by creating a synthetic control group that is as close as possible to the program group. Classical matching methods and pooled probit models rely on the assumption of conditional independence, i.e., based on a vector of observable covariates, the treatment is independent of unobservable factors.

Moreover, entropy balancing allows to account for the panel dimension of the data by controlling for country- and time-specific factors in the second stage of the regression analysis. It is necessary to include country-specific effects to account for potential unobserved heterogeneity between countries that haven't signed resource-backed loans and those that have signed resource-backed loans. Indeed, the macroeconomic environment of these two groups may differ beyond the covariates used in the entropy balancing approach. Country effects also control for time-invariant, country-specific conditions that may lead to differences in sovereign debt default across countries. Finally, entropy balancing may fail to control for potential endogeneity biases due to unobserved time-varying factors that may affect both resource-backed loans and sovereign debt default, as well as the reverse causality problem that may exist between the treatment variable and the outcome variable, and cannot successfully address the inertia of environmental sustainability. This can lead to potentially biased results. For this reason, We complement entropy balancing with alternative estimation methods such as Propensity Score Matching (PSM), Propensity score weighting (PSW), Coarsened exact matching (CEM) Panel fixed-effects and two-step system-GMM dynamic panel estimator (GMM).

4 Data

4.1 Variables description

Our study is covering the period 1990-2020. The dataset used in this study consists of 134 developing countries in which 14 are RBLs (treatment group) and 120 are non RBLs (control group). The time horizon of our study is motivated by the availability of data but also by the first and last resource-backed loan periods for which we have information. The year 1990's, for example, is chosen as the beginning of the study due to the paucity of budgetary data before this date but also because it corresponds to the first wave of RBL's negotiation with China[4].

Treatment variable : our treatment variable is resource-backed loan (RBL). We use information about the existence of a resource-backed loan from Natural Resource Governance Institute (NRGI) dataset. To date, NRGI remains the only institution that provides reliable and solid evidence on the existence and many details of resource-backed loans in developing countries. Indeed, a team of researchers led by Mihalyi, D., Adam, A. and Hwang, J. collected as much information as possible on resource-backed loans to obtain a somewhat larger database. They compiled two original dataset maintained by institutions dedicated to researching China's overseas activities: the China-Africa loans dataset produced by the China-Africa Research Initiative (CARI) at Johns Hopkins SAIS and the China-Latin America financial database of the Inter-American Dialogue and Global Development Policy Center at Boston University (Mihalyi et al., 2020). Next, the authors supplemented these databases with additional literature searches to present their dataset.

[4] Mihalyi et al. (2020) argue that the practice of negotiating and signing RBLs with Peru began in the 19th century and with Angola in the mid-1990s, when Chinese banks provided funds to the Angolan government at the time of the civil war and used future oil revenues to guarantee repayment.

It should be noted that the data on these resource-backed loans are not fully comprehensive due to the relative opacity of these transactions (Rivetti, 2021). There could certainly be other resource-backed loans in both regions than those listed by the authors as they are limited by publicly available information (Mihalyi et al., 2020). However, it is the most comprehensive dataset on resource-backed loans for African and Latin America economies over a long period. Based on literature (Mihalyi et al., 2020; 2022; Bräutigam and Gallagher, 2014), we construct a binary variable that takes the value 1 if country i at a period t has taken out a resource-backed loan and 0 otherwise.

Outcome variable : dependent variable used in this study is the face value of sovereign government debt in default. The data are collected from the defaulted public debt database developed by the Bank of Canada's Credit Rating Assessment Group (CRAG)[5]. To obtain the BoC–BoE sovereign default database, Bank of Canada researchers compile a comprehensive set of global data on public debt, defaults, and the stock of arrears to official creditors, using a variety of sources, including international and regional organizations (i.e., Asian Development Bank, IMF, Paris Club, World Bank, and IBRD financial statements) and academic authors (see, for instance, Beers et al., 2020). The BoC–BoE sovereign default database tabulates data on debt owed to official and private creditors for all sovereign defaults identified between the years 1960 and 2021. Although this database suffers from the shortcoming that very little effort is made to systematically measure and aggregate the nominal value of different types of defaulted sovereign debt (Beers et al., 2020), it is arguably the most comprehensive dataset on defaulted public debt in developing countries over a long period (Balima and Sy, 2021). Finally, we use this database for two other reasons. First, this database is relatively exhaustive with a frequency of occurrence of sovereign defaults, compared to the database of Reinhart and Rogoff (2009). Second, this basis is used by many authors in their recent studies (see, for instance, Balima and Sy, 2021; Reusens and Croux, 2017; Eichengreen, 2016; Reinhart and Trebesch, 2016; etc).

Control variables : regarding the control variables, our baseline regressions include similar covariates as Gomez-Gonzalez et al. (2022); Balima and Sy (2021); Eichler and Maltritz (2013); Hilscher and Nosbusch (2010); Forslund et al. (2011). Indeed, we will introduce a battery of additional covariates to ensure that the result is not driven by a specific choice of covariates in robustness checks. Our baseline covariates consist of the following: inflation rate, the GDP growth per capita, the natural resource rents, the fiscal balance, the public debt, domestic credit to private sector, and the broad money growth. Details of variable definitions and sources can be found in Table 29 in the Appendix.

4.2 Descriptive statistics and covariate balancing

We begin the discussion of descriptive statistics by assessing the relationship between resource-backed loans and sovereign debt default. To do so, we analyze and compare the

[5]The latest version of the database is provided by Beers and Mavalwalla (2017).

unconditional and conditional probabilities of default occurring in our sample. In Table 1, the unconditional probability of a sovereign default-which is the number of crises divided by the number of non-missing country-year observations -is 2.2 percentage points. However, the conditional probability-measuring the probability of a sovereign debt default if the country signed a resource-backed loan during the period of our study-is 3.7 percentage points.

Table 1: Conditional and unconditional probabilities of a debt default crisis

	Conditional	Unconditional	Difference
Sovereign debt default	0.037	0.022	0.015

Note: in this table, we present the conditional and the unconditional probabilities of the occurrence of a sovereign debt default crisis in our sample. The conditional probability is defined as the probability of experiencing a default, conditional on having signed at least an resource-backed loan. The unconditional probability is the number of crises divided by the number of non-missing country-year observations.

These correlations suggest that countries that have signed a resource-backed loan have about 1.5 percentage points more likely to experience a sovereign debt default crisis than the non-resource-backed countries in our sample.

Finally, we test the performance of entropy balancing for our covariates. To this end, we present in Tables 2 some descriptive statistics obtained before and after weighting to estimate the treatment effect of resource-backed loan adoption. Columns [1]-[2] of Panel A show the sample means before weighting for the country-year observations for the treatment — group (with resource-backed loans) and the control group (without resource-backed loans), respectively. Column [3] of this table shows the difference in means between the two groups. The results reveal a difference — between these two — groups. Indeed, resource-backed loans countries are characterized by high inflation rate, low GDP per capita, high natural resource rents, high broad money (M2), high public debt, and low domestic credit to the private sector. These results are mostly consistent with the expected relationship between the probability of signing resource-backed loans and the various reliable control variables discussed above. Columns [1] and [2] of Panel B show the sample mean after weighting between the treatment group and the synthetic group obtained by entropy balancing, and Column [3] shows the difference between the first two. The analysis of the two groups in this table reveals the effectiveness of entropy balancing, as the difference shown in the previous table seems to disappear. Thus, using entropy balancing, we can construct a perfect control group that is very similar to the resource-backed loan countries in terms of the means of the prepossessed covariates. Building on these correlations (see Table 1) and these statistics (see Tables 2 and 3), we therefore dig deeper into the analysis in the next section.

Table 2: Descriptive statistics and covariate balancing

Panel A : Descriptive statistics before weighting	***RBL***	***Non-RBL***	***Diff***
	[1]	[2]	[3]=[2]-[1]
Lag Inflation rate	11.69	7.296	-4.394***
Lag GDP growth per capita	1.168	2.353	1.185**
Lag Natural resource rents	15.3	6.661	-8.639***
Lag Fiscal balance	-3.326	-2.275	1.051**
Lag Public debt	58.78	55.86	-2.92
Lag Domestic credit to private sector	19.28	33.58	14.3***
Lag Broad money growth	22.23	14.93	-7.3***
Panel B : Descriptive statistics After weighting			
	RBL	***Non-RBL***	***Diff***
	[1]	[2]	[3]=[2]-[1]
Lag Inflation rate	11.69	11.69	0.00
Lag GDP growth per capita	1.168	1.168	0.00
Lag Natural resource rents	15.3	15.3	0.00
Lag Fiscal balance	-3.326	-3.326	0.00
Lag Public debt	58.78	58.78	0.00
Lag Domestic credit to private sector	19.28	19.3	0.02
Lag Broad money growth	22.23	22.23	0.00

***p < 0.01, **p < 0.05, *p < 0.1

Table 3: Covariate balancing

	RBL	***Non-RBL***	***Diff***	***t-TEST***	***P-value***
	[1]	[2]	[3]=[2]-[1]	[4]	[5]
Lag Inflation rate	11.69	7.296	-4.394	8.7	0.003
Lag GDP growth per capita	1.168	2.353	1.185	6.2	0.013
Lag Natural resource rents	15.3	6.661	-8.639	97.0	0.000
Lag Fiscal balance	-3.326	-2.275	1.051	5.3	0.021
Lag Public debt	58.78	55.86	-2.92	0.5	0.473
Lag Domestic credit to private sector	19.28	33.58	14.3	37.5	0.000
Lag Broad money growth	22.23	14.93	-7.3	23.9	0.000

This table reports in Column [1] the pre-weighted sample means of the matching covariates for the country-year observations in which a resource-backed loan was signed (the treatment group) and in Column [2] the country-year observations in which a resource-backed loan was not signed (the potential control group). Column [3] contains the mean differences between the treated and control groups, as well as the corresponding t-test statistics in Column [4] and p-values in Column [5].

5 Results

We present in Table 4 the impact of resource-backed loans on the sovereign debt default crisis using entropy balancing. First, before turning to the estimation of the treatment effect, we analyze the performance of entropy balancing in constructing a counterfactual fairly close to the units without resource-backed loans. Tables 2 and 3 present the sample means of the matching covariates before and after the weighting used to estimate the impact of resource-backed loans on sovereign debt default. We already notice in Table 2 that countries that have signed at least one resource-backed loan (Column [1]) differ from countries that have not (Column [2]). Indeed, countries that have signed at least one resource-backed loan have (a) high inflation rate, (b) low GDP per capita, (c) high share of resource rents, (d) high budget deficit, (e) high level of indebtedness, (f) low domestic credit to the private sector , and (g) high broad money (M2) compared to the other countries. However, after creating the balanced sample using the covariate moments, the results in Table 3 clearly show no significant difference between the two groups. The lack of differences between the groups strongly demonstrates the effectiveness of the entropy balancing method in establishing the desired equilibrium. The main conclusion is presented in Table 4. Column [1] shows the regression result without the corresponding covariates in the second stage of entropy balancing. Column [2] incorporates the covariates into the regression. Columns [3] and [4] control for year- and region-fixed effects, respectively. Finally, Column [5] brings together the covariates and the year- and region-fixed effects in the regression. Our results are robust: regardless of the specifications, the estimated effects of resources-backed loans on sovereign debt default are positive and statistically significant. The magnitude of the coefficients found ranges from 0.86 to 1.86 percentage points. The magnitude of the coefficient for our preferred specification is 1.49 percentage points (see Table 4, Column [5]). Our results suggest that resource-backed loans significantly increase sovereign debt default by 1.49 percentage points in countries with a resource-backed loan arrangement compared with non-resource-backed loan countries. This result is economically meaningful as it corresponds to about $1/49 \simeq 0.020$ of the unconditional probability of experiencing sovereign debt default in our sample.

Table 4: Resource-backed loans and sovereign debt default—Baseline results

VARIABLES	[1]	[2]	[3]	[4]	[5]
Resource-backed loans	1.8487***	1.8613***	0.8607***	1.7745***	1.4892***
	(0.2642)	(0.2096)	(0.2102)	(0.2497)	(0.2240)
Observations	2,250	2,250	2,250	2,250	2,250
R-squared	0.0927	0.3842	0.7798	0.3952	0.8000
Covariates in the second step	No	Yes	Yes	Yes	Yes
Country/FE in the second step	No	No	Yes	No	Yes
Year/FE in the second step	No	No	No	Yes	Yes

This table presents the effect of resource-backed loans on sovereign debt default obtained by weighted least squares regressions. The treatment variable is the presence of a resource-backed loan. The outcome variable is the occurrence of sovereign debt default. The control variables include GDP growth per capita, inflation rate, natural resource rents, fiscal balance, public debt, primary balance, and broad money. Column [1] reports the result without matching covariates in the second step of entropy balancing. Column [2] brings the covariates to the regression. Columns [3] and [4] control for year and regional fixed-effects, respectively. Finally, Column [5] gathers the covariates and year and regional fixed-effects into the second step regression. Robust standard errors are in parentheses ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

6 Robustness Checks

Our previous results show that the presence of a resource-backed loan increases sovereign debt default in signatory countries, relative to non-signatory countries. In what following, we perform a battery of robustness checks to ensure that this result is not sensitive to alternative specifications and identification strategies. In performing these robustness checks, we focus our attention on the sign and statistical significance of the main variable of interest — resource-backed loans.

6.1 Alternative definition of sovereign debt default

In this subsection, we examine the robustness of the main result of the paper by changing our dependent variable. To do so, we follow Balima and Sy (2021) and Jorra (2012) which both use a dummy variable to capture the sovereign debt default. Indeed, Jorra (2012) following Celasun and Harms (2011); Kohlscheen (2010) and Van Rijckeghem and Weder (2009), defines sovereign debt default as the actual impact of sovereign defaults on foreign currency debt as reported by the Standard & Poor's rating agency[6]. While Balima and Sy (2021) construct a dummy variable equal to 1 for country-year observations in which there was a sovereign default and zero otherwise. Based on these two authors, we construct

[6]According to Standard and Poor's (2009), sovereign debt default is defined as the failure to pay the principal or interest on sovereign debt by the maturity date (or within the specified grace period) specified in the original terms of a debt issue. Thus, it excludes defaults with official creditors (see Balima and Sy (2021) and Standard and Poor's (2006)).

a dummy variable that has the value 1 if country i at period t had a sovereign debt default crisis and 0 otherwise according to each author's definition. The main results of the paper remain qualitatively unchanged when another definition is used to measure sovereign debt default. The estimation results suggest that resource-backed loans increase sovereign debt default in signatory countries than in non-signatory countries under both definitions (Table 5, Column 1 and 2).

Table 5: Resource-backed loans and sovereign debt default—Robustness checks

	[1]	[2]
VARIABLES	Balima & Sy's definition	Jorra's definition
Resource-backed loans	0.0344**	0.0069*
	(0.0172)	(0.0140)
Observations	2,277	2,277
R-squared	0.6494	0.1528
Covariates in the second step	Yes	Yes
Country/FE in the second step	Yes	Yes
Year/FE in the second step	Yes	Yes

This table presents the effect of resource-backed loans on sovereign debt default obtained by weighted least squares regressions. The treatment variable is the presence of resource-backed loan. The outcome variable is a sovereign debt default. The control variables include GDP growth per capita, inflation rate, natural resource rents, fiscal balance, public debt, and primary balance, and Broad money. Column [1] rely on Balima and Sy (2021)'s definition of sovereign debt default while column [2] reports Jorra (2012)'s definition of debt crisis. Robust standard errors are in parentheses ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

6.2 Alternative specifications

6.2.1 Modification of sample

Previous work on sovereign defaults has shown that episodes of large-scale sovereign debt overhang and default are generally associated with a sustained period of sub-par growth (see, Reinhart and Trebesch, 2016; Reinhart and Rogoff, 2013; Reinhart et al., 2012; Reinhart and Rogoff, 2011; Reinhart et al., 2003). For example, Reinhart and Rogoff (2013) show that banking crises weaken fiscal positions, invariably contract government revenues, and increase central government debt by about 86% three years after a crisis. The coronavirus pandemic 2019 has driven up inflation, which has been exacerbated by the conflict in Ukraine, adding to the economic pain (Nations, 2016). Interest rate hikes in response to this inflation have made it even more difficult for many countries to make new loans and have increased interest rates on existing debt. As a result, more than 150 countries are expected to cut public spending and tighten austerity measures in 2021 and

2022 to free up funds needed to stabilize debt levels and meet debt service obligations (Ortiz and Cummins, 2021).

Failure to take these observations into account could bias the conclusion of our results in Table 4. We therefore take these considerations into account by first modifying our overall sample. To do this, we exclude certain countries or observations from our sample. The results of the regressions are presented in Table 6. In Columns [1]-[2], we exclude the year-observations where there were currency and banking crises respectively. Following Reinhart and Rogoff (2011), we understand that a decline in GDP or the likelihood of a prolonged period of below-average growth generally follows from a sovereign debt default. At the same time, an implosion of growth has as a counterpart an explosion of the public debt (Easterly, 2001). This could trigger a sovereign debt default. We take this into account by excluding the yearly observations with a weighted GDP closer to or equal to zero (Column [3]). In column [4], we exclude countries where a resource-backed loan has recently been signed because of the potential lag between the reforms and their expected effects. In Columns [5]-[6], we consider the effect of sub-prime and oil crises on over-indebtedness and sovereign debt default. Many of the countries in our sample were over-leveraged after the 2014 oil price drop (see, Mihalyi et al., 2020). In columns [7]-[8], we explore the direct or indirect ways in which inflation or currency depreciation can also erode the value of certain types of debt. We define an inflation crisis using a threshold of 20% of the inflation rate per year. Hyperinflation defined as episodes in which the annual inflation rate exceeds 500% of the inflation rate in modern times (see, Reinhart and Rogoff, 2004). Finally, we delete the observations-years of the period of the coronavirus pandemic in Column [9]. The main results of the document remain qualitatively unchanged when we modify our overall sample. Resource-backed loans have a positive and statistically significant impact on sovereign debt default (Columns [1]-[9]).

Table 6: Resource-backed loans and sovereign debt default—Robustness results: Excluding year-observations

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
VARIABLES	Currency crises	Banking crisis	GDP weighted default	New loans	Economic crisis	Petroleum crisis	Hyperinflation years	Inflation Crises	Covid-19 crisis
Resource-backed loans	1.3935***	1.3237***	1.1646***	1.4754***	1.5545***	1.4407***	1.2149***	1.3105***	1.4714***
	(0.2224)	(0.2152)	(0.2150)	(0.2799)	(0.2257)	(0.2261)	(0.2211)	(0.2160)	(0.2303)
Observations	2,175	2,200	1,689	1,708	2,152	2,137	2,204	2,216	2,143
R-squared	0.8204	0.8212	0.8610	0.8103	0.8018	0.8059	0.7947	0.8155	0.7993
Covariates in second step	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE in second step	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year/FE in second step	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Undeclared constant included. Standard errors in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Note: Observed differences between the observations in Table 4 and those in this table are due to the number of observations removed.

6.2.2 Adding control variables

After modifying the sample, we also consider increasing the specificity of the baseline model to test the robustness of our results. To do this, we control across several additional variables that are likely to be positively or negatively effect with sovereign debt default. Based on the literature, we control for the effect of a few variables: financial openness, government expenditure, Interest on debt, government revenue, financial development, crude oil brent, and commodity prices (Net barter terms of trade index and terms of trade index). According to Balteanu and Erce (2018), financially more open countries prior to a crisis are more likely to experience twin bank debt crises, and less likely to experience single and sovereign debt crises. In contrast, countries in the process of opening up are more likely to experience both single and twin bank debt crises. We take this into account by controlling financial openness. Next, we control the effect of interest rates on debt. Indeed, it is well known that in most cases, borrowing decisions are influenced by the level and volatility of global interest rates. Johri et al. (2022) find that debt issuance decreases when the world interest rate rises, and that this reactivity is higher in the high-volatility state than in the low-volatility state. Temporal variation in the global interest rate interacts with default incentives, and its effect on sovereign borrowing and spreads depends on the state (Johri et al., 2022).

The literature indicates that a tighter fiscal policy reduces the amount of borrowing required to repay maturing debt, making the government less likely to default in the future (Liu and Shen, 2022). To this end, we control for the effect of government expenditure and government revenue. According to Bulow and Rogoff (1989), changes in a country's terms of trade affect its ability to generate dollar-denominated income from exports, and therefore its ability to make payments on its dollar-denominated external debt. In fact, terms-of-trade volatility accounts for the variability of output at business cycle frequencies in general (Mendoza, 1995), and has negative effects on long-term growth (Mendoza, 1997). Since resource-backed loans are generally taken out in US dollars, collapse in natural resources prices could increase the nominal sizes of these loans (Mihalyi et al., 2022) and thereby lead borrowing countries to sovereign debt default risk. We take these factors into account by controlling crude oil brent and commodity prices. For the reason of reverse causality, these variables included essentially in the second stage of the estimation are all lagged by one period. Columns [1]-[8] of Table 8 and 9 in Appendix B, which report the results of these specifications, show their consistency with our baseline findings (see Table 4). In other words, adding these additional covariates does not change our results. In addition, given that any characteristics able to predict resource-backed loans can be a source of endogeneity, we augment the variables used in the baseline first-stage model by adding these additional control variables to construct the synthetic control. The results in Table 10 (Columns [1]-[8]) in Appendix B show that adding these variables to the determinants of resource-backed loans does not change our results. Resource-backed loans increases sovereign debt default.

6.2.3 Alternative estimation method

We perform another robustness test by changing my identification strategy. We use three alternative methods: propensity score matching, propensity score weighting, coarsened exact matching, Balancing of higher moments, panel fixed-effects (Driscoll-Kraay), and system GMM.

Propensity score matching (PSM): According to Rosenbaum and Rubin (1983), the propensity score matching estimation consists of two steps. First, we calculate the likelihood for a country signing resource-backed loans, conditional on the covariates used in the baseline model. In a second step, the propensity scores obtained are used to match treated and untreated observations, then the ATTs are computed to estimate the effect of the treatment. In line with the existing literature (see, Rosenbaum and Rubin, 1983, Dehejia and Wahba, 2002, Smith and Todd, 2005, Heckman et al., 1998), we draw upon four propensity score matching methods to pair up treated with comparable untreated observations : the N-nearest-Neighbors method (***N=1, N=2, and N=3***); the radius method (***r = 0.005; r=0.01; and r=0.05***, respectively); the Kernel method and the Local Linear Regression (see, Heckman et al., 1998). Moreover, we impose the common support, which allows matching each treated observation with at least one untreated counterfactual that is as similar as possible. New ATTs are reported in Table 11 in appendix, with the tests relating to the quality of the matching. First, results are stable. Second, all the Pseudo-R2 in our estimates are less than 10%, suggesting that the matching provided balanced scores. That is, our findings are robust regarding the hypothesis of common support. Finally, our findings are also robust regarding the Conditional Independence Assumption (CIA), since the cutting points from Rosenbaum sensitivity tests at 10% significance hover between 1.95 and 3.0, comparable with existing studies (see for instance, Rosenbaum and Rosenbaum, 2002 and Aakvik et al., 2001). Therefore, new findings strengthen our main results.

Propensity score weighting (PSW): Propensity score weighting assigns individuals different "weights"—weighting them up or down to make the individuals in the treatment group and the comparison group more similar to each other. One of the advantages of using propensity score weighting, as opposed to matching, is that you're able to include all individuals; none of the individuals are excluded because they can't be matched to a individual in the other treatment arm. Indeed, including all the units is especially important when you have small sample sizes. Propensity score weighting allows to leverage information from all units included in our sample. Propensity score weighting is frequently used in a variety of contexts, and recent studies have estimated causal effects (see, for instance, Hirano and Imbens, 2001 and Lee et al., 2011). Results reported in Columns ([1]-[5]) of Table 12 suggest a positive and significant effect of resource-backed loans on sovereign debt default, with a magnitude of about 1.6 percentage points, qualitatively comparable to that of the main model (1.5 percentage points). For greater robustness, we ran the regressions using the ***propensity score weighting*** method. However, we have taken care not to present this method here. The results are shown in Table 24 in the appendix. All the ATTs resulting from this regression remain significant, confirming the robustness of our main results.

Coarsened exact matching (CEM) : By definition, coarsened exact matching is a monotonic matching method that reduces imbalances, meaning that the balance between the treatment and control groups is chosen by the user ex ante, rather than being discovered through the usual laborious process of post-hoc verification, method adjustment and repeated re-estimation (Iacus et al., 2012; Blackwell et al., 2009; Iacus et al., 2009). The advantage of this method is that it first sorts all observations into strata, each with identical values for all covariates before treatment, and then discards all observations within a stratum that do not have at least one observation for each unique value of the treatment variable (see, Iacus et al., 2012). According to Iacus et al. (2009), CEM bounds both the error in estimating the average treatment effect and the amount of model dependence. Finally, CEM has a number of other beneficial properties. First, CEM meets the congruence principle, which states that the data space and analysis space should be the same. Methods that fail to meet this principle often produce strange or counterintuitive results. Methods that meet the principle allow analysts to leverage their substantive knowledge of the data to find better matches. Second, CEM automatically restricts the matched data to areas of common empirical support. This is necessary to remove the possibility of difficult-to-justify extrapolations of the causal effect that end up being heavily model dependent (King and Zeng, 2006). Table 13 reports the regression results for each strata specification. Panel A shows the results for CEM matched, Panel B shows those for CEM strata, while Panel C displays the results for CEM weights. This new results lead to qualitatively similar conclusions to the baseline results.

Balancing of higher moments (variance): The goal of entropy balancing is to find a vector of weights that balances the data between two subsamples with respect to specific moments of a list of variables (Hainmueller, 2012). If nothing is specified, by default, only the means of the specified variables will be balanced. The objective of balancing of higher moments is to compensate for the shortcomings of the mean by specifying the variance (Hainmueller, 2012). The results are shown in Tables 14 and 15. The new results are similar to the baseline results.

Panel fixed-effects (OLS estimates): We test the robustness of our results using the Ordinary Least Squares (OLS) or a panel fixed-effects regression (Driscoll-Kraay). Results reported in Columns ([1]-[2]) of Table 16 suggest a positive and significant effect of resource-backed loans on sovereign debt default, with a magnitude of about 1.5 percentage points, qualitatively comparable to that of the main model (1.4 percentage points).

Generalized Moment Method (GMM): Finally, we re-estimate main model following Blundell and Bond (1998), using two-step system-GMM dynamic panel estimator. Our results may suffer from endogeneity bias. In fact, a problem of reverse causality may exist between resource-backed loans and sovereign debt default in the sense that certain countries that have restructured their debt in several series with sovereign debt defaults may be tempted to sign a resource-backed loan. Our resource-backed loans variable — is therefore suspected to be endogenous. Moreover, countries that frequently default on their debt experience episodes of high inflation or poor macroeconomic performance (Sunder-Plassmann, 2020). Hence the importance of using the GMM estimator to correct this problem. This method allows controlling for the persistence of fiscal outcomes, notably sovereign debt default, to control for the Nickell bias (Nickell, 1981) that arises in a dynamic panel model — a model with the lagged endogenous in the control variables — with fixed effects, to limit the influence of time-varying unobservable factors that may influence both the outcome (sovereign debt default) and treatment variable (resource-backed loans), and to mitigate the challenge of finding an exogenous instrument for estimating the effect of the resource-backed loans. The results presented in Table 17 lead to qualitatively similar conclusions to the baseline results. We have verified all the validity conditions of the GMM estimator. First, regarding the instrument selection criteria, the Hansen test does not reject the hypothesis of instrument validity. Besides that, we need to ensure that the total number of instruments does not exceed the number of countries, to avoid the problem of ***"instrument proliferation"*** in estimates (Roodman, 2009). Finally, the AR (1) test for the absence of autocorrelation of the first-order error term and the AR (2) test for the absence of autocorrelation of the second-order error term do not raise concerns about the validity of our estimates.

6.3 Threats to identification : Falsification tests

So far, the various robustness tests carried out confirm our conclusions. However, it may turn out that the adoption of a resource-backed loan is generally associated with parallel reforms. It is therefore conceivable that unobservable variables correlated with the adoption of resource-backed loans and potentially with sovereign debt default could influence our results. A related concern is that there could certainly be other resource-backed loans in developing countries than those listed by Mihalyi et al. (2020), given that they are limited by publicly available information (Mihalyi et al., 2020, 2022; Horn et al., 2021; Rivetti, 2021). We therefore fear that the control group may be contaminated. Admittedly, it is recognized in the literature that the empirical method used in this study aims to address this type of concern (see, Hainmueller, 2012). Nevertheless, we reinforce our results by performing a few additional tests in this subsection. First, in Columns ([1]-[5]) of Table 17, we perform a random date assignment of resource-backed loan signatures to the treated countries. Secondly, we perform a random assignment to the treatment in the whole sample, considering respectively 10%, 20%, 30%, 40% , 50%, and 60% of the sample observations as treated in Table 19 (Columns [1]-[6]). If our results are biased by unobservables or a trend, placebo tests could also show significant effects. Indeed, random treatments within the sample have no impact on sovereign debt default. Consequently, we can exclude the possibility that confounding factors or trends influence our results.

6.4 Resource-backed loans effect: The time perspective

The effects obtained so far from the signature of a resource-backed loan are a comparison of conditional efficiency averages in periods when the resource-backed loans have been signed versus periods when they are not. The effect captured in this paper may suffer from some problems. In fact, adopting resource-backed loans can trigger a change in the economic (Investment), political (conflicts), institutional (corruption), and social environment of countries. In this sense, it can be argued that the effect captured may not be due to the resource-backed loans but to changes in institutional, political, social, or economic conditions after its adoption. Similarly, any other characteristic that may determine the adoption of resource-backed loans, but not included in the econometric specification, may be a source of endogeneity. To circumvent these problems, it would be interesting to compare the results obtained from the full sample period with those around a smaller period. A narrower sample window should provide a more robust estimate of the effects of the resource-backed loan, since confounding factors such as the implementation of other reforms or change in political regime are more likely to play out over time (see, for instance, Apeti, 2023; Neuenkirch and Neumeier, 2015). That is, we deepen the robustness of our results with four modifications of our initial sample, considering a window of two, three, four and five years, respectively. Results are presented in Table 20, Columns ([1]-[4]). Overall, the coefficients obtained from a smaller window are around those obtained from the full sample. Therefore, it seems unlikely that the estimated effect of resource-backed loans is due to a fortuitous change in the policy or institutional environment of the treated country.

7 Channels

This section empirically tests the main channels through which the causal effects of financial market access[7] and quality of institution[8] may operate which are our two transmission channels.

7.1 Do borrowing costs amplify sovereign debt defaults?

The first channel we are testing is the cost of borrowing on the financial market. Indeed, one of the main reasons for resorting to resource-backed loans is that many developing countries have limited access to the main credit markets due to their history of default. So, when a country uses a guarantee such as resource-backed loans to obtain financing funds, it sends out two different types of signal regarding its ability to repay the loan on time. The first signal is the negative perception that the borrowing country gives backers about repayment. In fact, if the borrower uses its natural resources as collateral, this

[7]For example, see, Sawadogo (2020).

[8]Captured by corruption (see, for instance, Mihalyi et al. (2022)).

could lead donors to believe that the borrower does not have a good financial reputation and will not be able to meet its repayment commitments. As a result, a loan is granted, but at a very high cost. The second signal is rather the assurance that the borrower gives to lenders with its natural resources. No doubt the lender will be able to lend him funds at a fairly reasonable cost through a breach of trust. In both cases, the borrowing country accesses the financial markets, but compromises the viability of its debt through the cost of borrowing. Hence the need to test this channel. As is well known, explicit and singular methods for evaluating transmission channels are not found in the literature. For this reason, we decided to stick with our basic entropy balancing method for testing these channels. The results presented in Table 21, Columns ([1]-[2]) show that signing resource-backed loans increases borrowing costs and public debt viability.

7.2 Do resource-backed loans lead to destabilization of fiscal discipline through corruption?

The second channel we are testing is the institutional quality captured by corruption. Our intuition is that the opacity of resource-backed loans allows corruption to flourish, and corruption reduces the revenue available to repay debt, resource-backed loans increase the amount of debt to be repaid through untracked loans. Ultimately, corruption and unpaid loans reinforce each other, leading to over-indebtedness. Results presented in Table 21, Column [3] in the Appendix show that the adoption of resource-backed loans increases corruption by 1.03 percentage points. These results validate our two main channels and show that the destabilizing effect of resource-backed loans on sovereign debt default can be fuelled by borrowing costs and weak institutional quality, in particular corruption.

8 Heterogeneity

8.1 Resource-backed loans disaggregated

Using the NRGI database developed by Mihalyi et al. (2020), we disaggregate resource-backed loans dummy into three types of resources used: oil resource-backed loans, minerals resource-backed loans, and other resource-backed loans. The aim of this exercise is to test the hypothesis that the probability of sovereign debt default will react differently depending on the type of resources considered for repayment. Already, the composition of resource-backed loans shows that 43 of the 52 loans signed were oil-backed, 9 were minerals-backed, and the remaining 3 were other-backed (Mihalyi et al., 2020). According to Coulibaly (2023), there is considerable heterogeneity between the loans signed, with minerals, tobacco and cocoa accounting for around 55% of total loans as a percentage of GDP, while petroleum accounted for around 45%. We are testing whether this heterogeneity has a major impact on sovereign debt default. The results in Table 22 present the conclusions of this hypothesis. We find that signing resource-backed loans increases sovereign debt default regardless of the type of resource used for repayment. However, we observe a relative variation in coefficients according to the type of resource-backed loan, corroborating our intuition.

8.2 Does EITI implementing matter for country signatory?

In this subsection, we test whether the implementation of the Extractive Industries Transparency Initiative (EITI) can mitigate the positive effect of resource-backed loans on sovereign debt default. In fact, the idea behind EITI is to promote understanding of natural resource management, strengthen public and corporate governance and accountability, and provide the data needed for policy-making and multi-stakeholder dialogue in the extractive sector. In addition, the EITI provides data that can help identify and close down channels of corruption in the mining, oil and gas sectors. However, its effect is not sufficiently clear in the literature. Indeed, some studies have concluded that, EITI contributes to preserving the forest cover (Bakehe, 2020; Kinda and Thiombiano, 2024) and improving the investment climate (Sovacool, 2020), economic development (Corrigan, 2017), governance reform (Arond et al., 2019), mitigating corruption (Villar, 2022) and building accountability (Villar, 2020) in the governance of extractive resources. In contrast to these authors, some believe that EITI membership has not had a positive impact on natural resource management (Sovacool et al., 2016; Kasekende et al., 2016). Our intuition is based on the fact that if resource-based lending lacks transparency, as emphasized by Mihalyi et al. (2020); Horn et al. (2021); Rivetti (2021); etc., EITI membership could help adopting countries to take advantage of and circumvent fiscal discipline problems.

Table 7: The Impact of EITI Membership

	[1]	[2]
VARIABLES	EITI countries	non-EITI countries
Resource-backed lonas dummy	-0.7184***	0.6371***
	(0.2066)	(0.2117)
Constant	8.3743***	7.2508***
	(0.3736)	(0.3638)
Covariates	No	No
Country/FE	Yes	Yes
Year/FE	Yes	Yes
Observations	184	1,675
R-squared	0.8759	0.8439

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

The results presented in Table 7 illustrate our hypothesis. Resource-backed loans are more efficient and reduce sovereign debt defaults in EITI member countries.

8.3 The contribution of structural factors

Despite certain common characteristics, developing countries are subject to a number of important structural differences that shape the effect of resource-backed loans on sovereign debt default. Consequently, this sub-section examines the heterogeneity effect of resource-backed loans according to two characteristics: ***i-Macroeconomic factors***: Government Expenditure volatility, inflation targeting, FDI, remittance inflows, and resource depletion; ***ii-Political and institutional factors***: Democracy, rule of low, and political stability. First, we analyze the effect of resource-backed loans concerning public spending volatility, inflation targeting, FDI, remittances and resource depletion. Given their link with sovereign debt default (Gong and Qian, 2022; Beers et al., 2020; Reinhart and Rogoff, 2009; Reinhart and Trebesch, 2016; Reusens and Croux, 2017; Fuentes and Saravia, 2010; Minea and Tapsoba, 2014; Balima et al., 2017), we hypothesize that the effect of resource-backed loans may be mitigated in situations characterized by high public expenditure volatility, in the presence of inflation targeting, high FDI inflows, high remittances, and resource depletion. To test this hypothesis, we include an interaction between each of these variables and resource-backed loans. Accordingly, a statistically significant interaction means the presence of heterogeneity. Results in Columns ([1]-[5]) of Table 23 support our hypothesis. Indeed, with the exception of remittances inflows, we find a negative and significant interaction effect between resource-backed loans and the other variables, i.e., government expenditure volatility, inflation targeting, FDI, and resource depletion. These results suggest that the effect of resource-backed loans on sovereign debt default may be attenuated in situations characterized by high public expenditure volatility, in presence of inflation targeting, high FDI inflows, and resource depletion.

However, the interaction between resource-backed loans and remittance inflows is positive and significant. This means that, in the presence of high remittance inflows in the host country, resource-backed loans amplify sovereign debt default. While without interaction the results show that remittance inflows reduce sovereign debt default, these results corroborate those found by Balima and Combes (2019).

Secondly, we analyze the heterogeneous effect of resource-backed loans with political and institutional variables (in particular rule of law, democracy, and political stability). The results in Columns [6]-[7] show a greater effect of resource-backed loans in situations of good institutional conditions, i.e. a high level of rule of law, and good application of democracy, thus supporting the literature on the influence of institutional quality on policy effectiveness (Llanto and Gonzalez, 2007; Apeti, 2023). However, we find no significant effect with political stability even though the coefficients are negative.

In total, our results are sensitive to many structural characteristics. The effect of resource-backed loans adoption is unclear in some circumstances, mainly related to the macroeconomic conditions and the political situations.

9 Conclusions and policy implications

The volatility of conventional financing sources and fiscal pressures linked to economic recessions, combined with difficulties in accessing capital markets, have led some resource-rich African countries to resort to resource-backed loans to finance infrastructure projects in large part, but not exclusively. Although resource-backed loans represent a good opportunity to finance critical development needs, they can also weaken debt sustainability and plunge the signatory country into a sovereign debt default episode. This paper analyzes the effect of adopting resource-backed moans on sovereign debt default. Using a large sample of 134 developing countries over the period 1990-2020 and relying on entropy balancing, we show that countries signing up to resource-backed loans present on average a sovereign debt default of 1.42 percentage points compared to non-signing countries. This result is robust to various tests, including alternative specifications, alternative sovereign debt default measures including Balima and Sy (2021)'s definition and Jorra (2012)'s definition, and alternative estimation methods. Analysis of the transmission channels shows that borrowing costs and corruption are the main channels through which resource-backed loans pass to trigger a sovereign debt default in developing countries. However, results reveal some heterogeneity across the type of resource-backed loans, and structural factors such as government expenditure volatility, inflation targeting, FDI, remittance inflows, resource depletion, democracy, and rule of low. Additional results highlighted in this paper show that the effect of resource-backed loans on sovereign debt default as a function of time. The literature shows that resource-backed loans present potential risks (Mihalyi et al., 2022). The major contribution of this paper to the literature is that it not only confirms the debt viability risk found by Mihalyi et al. (2020), but also detects a new risk, sovereign debt default, which is caused by resource-backed loans and to which signatory countries may be exposed. This may be due to the fact that resource-backed loans are often negotiated in a non-transparent way, with opaque agreements concluded following non-competitive procedures, and are also carried out outside the general budget by state-owned companies with dubious management. Based on these findings, we recommend that policymakers in signatory and future signatory countries ensure full transparency of these loans, record the loans in the government budget, and invest productively so that productive investments are able to generate long-term returns that exceed their financing costs.

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Appendix Robustness results, Descriptive statistics, and Source of data used

Table 8: Resource-backed loans and debt default: additional controls in first stage

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
VARIABLES	Financial openness	Government expenditure	Interest on debt	Government revenue	Financial development	Crude oil brent	Terms of trade index IMF	Terms of trade index WDI
Resource-backed loans	0.9332***	0.8286***	0.9605***	0.8294***	0.4218*	0.7383***	0.9091***	1.4971***
	(0.2462)	(0.2644)	(0.2651)	(0.2631)	(0.2167)	(0.2261)	(0.2264)	(0.2415)
Observations	3,543	2,688	2,689	2,688	3,733	3,961	3,700	3,347
R-squared	0.6448	0.6153	0.6139	0.6157	0.6180	0.6268	0.6348	0.6807
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Undeclared constant included. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Note: The differences that can be observed between observations in Table 4 and those in this table are attributable to the number of observations of additional variables.

Table 9: Resource-backed loans and debt default: additional controls in second stage

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
VARIABLES	Financial openness	Government expenditure	Interest on debt	Government revenue	Financial development	Crude oil brent	Terms of trade index IMF	Terms of trade index WDI
Resource-backed loans	0.7231***	0.5918**	0.6978**	0.7909***	0.5925**	0.6683***	0.9071***	0.5058*
	(0.2604)	(0.2681)	(0.2934)	(0.2334)	(0.2681)	(0.2561)	(0.2605)	(0.2591)
Observations	2,569	2,170	2,170	2,652	2,170	2,653	2,523	2,707
R-squared	0.7644	0.7632	0.7602	0.7617	0.7632	0.7998	0.8232	0.8206
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Undeclared constant included. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Note: The differences that can be observed between observations in Table 4 and those in this table are attributable to the number of observations of additional variables.

Table 10: Resource-backed loans and debt default: additional controls in both second stage and first stage

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
VARIABLES	Financial openness	Government expenditure	Interest on debt	Government revenue	Financial development	Crude oil brent	Terms of trade index IMF	Terms of trade index WDI
Resource-backed loans	0.7290***	0.5942**	1.0298***	0.7990***	0.5693**	0.6620***	0.8983***	0.5058*
	(0.2624)	(0.2718)	(0.2620)	(0.2341)	(0.2723)	(0.2450)	(0.2566)	(0.2591)
Constant	-0.4794	5.9923***	5.1269***	0.5797	5.5342***	4.1333***	7.7645***	6.5838***
	(0.8107)	(0.9029)	(0.7642)	(1.3860)	(0.8955)	(1.3432)	(0.5770)	(0.7422)
Observations	2,569	2,170	2,170	2,652	2,170	2,653	2,523	2,707
R-squared	0.7644	0.7634	0.7709	0.7618	0.7633	0.8037	0.8287	0.8206
Country/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Undeclared constant included. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Note: The differences that can be observed between observations in Table 4 and those in this table are attributable to the number of observations of additional variables.

Table 11: Resource-backed loans and sovereign debt default: using Propensity Score Matching (PSM)

Dependent Variable :	1-Nearest Neighbor	2-Nearest Neighbor	3-Nearest Neighbor	Radius Matching			Local lineair regression	Kernel
Sovereign debt default		Matching		r=0.005	r=0.05	r=0.01	Matching	Matching
ATT	1.6720***	1.9393***	1.9856***	1.9778***	2.1073***	1.9570***	2.0125***	2.1012***
	(0.4922)	(0.3636)	(0.2961)	(0.2542)	(0.1875)	(0.2320)	(0.2204)	(0.1846)
Observations	2,250	2,250	2,250	2,250	2,250	2,250	2,250	2,250
Treated	2,122	2,122	2,122	2,122	2,122	2,122	2,122	2,122
Untreated	128	128	128	128	128	128	128	128
Pseudo-R2	0.022	0.013	0.010	0.004	0.012	0.002	0.022	0.010
Standardized biases (p-value)	0.340	0.692	0.833	0.989	0.767	0.999	0.340	0.831
Rosenbaum bounds sensitivity tests	1.95	2.7	3.0	3.0	3.0	3.0	3.0	3.0

Boot-
strapped standard errors based on 50 replications reported in brackets. *** p<0.01, ** p<0.05, * p<0.1. Note: No difference with observations in Table 4.

Table 12: Resource-backed loans and sovereign debt default: using PSW

VARIABLES	[1]	[2]	[3]	[4]	[5]
	Resource-backed loans on sovereign debt default				
ATT	2.0800***	1.8305***	0.9117***	1.8719***	1.6237***
	(0.3322)	(0.2946)	(0.2000)	(0.2808)	(0.2285)
Constant	3.2830***	4.0308***	5.8388***	3.9487***	6.7314***
	(0.0656)	(0.2894)	(0.2115)	(0.7195)	(0.6410)
Covariates	No	Yes	Yes	Yes	Yes
Country/FE	No	No	Yes	No	Yes
Year/FE	No	No	No	Yes	Yes
Observations	2,250	2,250	2,250	2,250	2,250
R-squared	0.1150	0.3686	0.7460	0.3859	0.7706

Notes : This Table presents the effect of resource-backed loans on sovereign debt default obtained by propensity score weighting (PSW). The treatment variable is resource-backed loan. The outcome variable is sovereign debt default. No difference with observations in Table 4. *** p<0.01, ** p<0.05, * p<0.1.

Table 13: Resource-backed loans and sovereign debt default: using CEM

VARIABLES	[1]	[2]	[3]	[4]	[5]
			PANEL A: CEM matched		
ATT	2.7818***	1.9188***	1.0058***	1.8758***	1.5239***
	(0.2483)	(0.2250)	(0.2869)	(0.2345)	(0.3148)
Constant	3.1560***	3.5462***	5.6546***	3.8574***	6.4942***
	(0.0756)	(0.2219)	(0.2436)	(0.7220)	(0.7298)
Observations	1,458	1,426	1,426	1,426	1,426
R-squared	0.0723	0.2672	0.7348	0.2852	0.7531
Covariates	No	Yes	Yes	Yes	Yes
Country/FE	No	No	Yes	No	Yes
Year/FE	No	No	No	Yes	Yes
VARIABLES	[1]	[2]	[3]	[4]	[5]
			PANEL B: CEM Strata		
ATT	2.5914***	1.7490***	0.9100***	1.8721***	1.4252***
	(0.2003)	(0.2149)	(0.2031)	(0.2243)	(0.2169)
Constant	3.4539***	3.3894***	5.8856***	3.5976***	6.8160***
	(0.0541)	(0.1518)	(0.2288)	(0.6767)	(0.6242)
Observations	4,045	2,222	2,222	2,222	2,222
R-squared	0.0311	0.3084	0.7466	0.3133	0.7630
Covariates	No	Yes	Yes	Yes	Yes
Country/FE	No	No	Yes	No	Yes
Year/FE	No	No	No	Yes	Yes
VARIABLES	[1]	[2]	[3]	[4]	[5]
			PANEL C: CEM Weigths		
ATT	1.9306***	1.8796***	1.1154***	2.0359***	1.9454***
	(0.2850)	(0.2453)	(0.2727)	(0.2666)	(0.3478)
Constant	4.0071***	4.1703***	5.7436***	6.2658***	8.4299***
	(0.1591)	(0.5381)	(0.2793)	(1.1744)	(0.9406)
Observations	1,458	1,426	1,426	1,426	1,426
R-squared	0.0360	0.2820	0.7040	0.3313	0.7495
Covariates	No	Yes	Yes	Yes	Yes
Country/FE	No	No	Yes	No	Yes
Year/FE	No	No	No	Yes	Yes

Notes : This Table presents the effect of resource-backed loans on sovereign debt default obtained by coarsened exact matching. The treatment variable is resource-backed loan. The outcome variable is sovereign debt default. No difference with observations in Table 4. *** p<0.01, ** p<0.05, * p<0.1.

Table 14: Resource-backed loans and sovereign debt default: using balancing of variance

VARIABLES	[1]	[2]	[3]	[4]	[5]
Resource-backed loans	1.5994***	1.6092***	0.8206***	1.4855***	1.3812***
	(0.2687)	(0.2083)	(0.1851)	(0.2445)	(0.2093)
Constant	4.4466***	4.2254***	5.7715***	3.9627***	6.3835***
	(0.1417)	(0.3226)	(0.2391)	(1.0588)	(0.9626)
Observations	2,222	2,222	2,222	2,222	2,222
R-squared	0.0741	0.3983	0.7943	0.4127	0.8087
Covariates	No	Yes	Yes	Yes	Yes
Country/FE	No	No	Yes	No	Yes
Year/FE	No	No	No	Yes	Yes

Notes : This Table presents the effect of resource-backed loans on sovereign debt default obtained by balancing of variance. The treatment variable is resource-backed loan. The outcome variable is sovereign debt default. No difference with observations in Table 4. *** p<0.01, ** p<0.05, * p<0.1.

Table 15: Resource-backed loans and sovereign debt default: using balancing of covariance

VARIABLES	[1]	[2]	[3]	[4]	[5]
Resource-backed loans	1.8484***	1.8531***	1.0742***	1.7766***	1.7023***
	(0.3062)	(0.2308)	(0.2325)	(0.2636)	(0.2686)
Constant	4.1976***	4.3761***	5.9281***	4.8611***	6.9037***
	(0.2040)	(0.3755)	(0.2884)	(1.3014)	(0.9187)
Observations	2,222	2,222	2,222	2,222	2,222
R-squared	0.0924	0.3810	0.7977	0.3980	0.8159
Covariates	No	Yes	Yes	Yes	Yes
Country/FE	No	No	Yes	No	Yes
Year/FE	No	No	No	Yes	Yes

Notes : This Table presents the effect of resource-backed loans on sovereign debt default obtained by balancing of covariance. The treatment variable is resource-backed loan. The outcome variable is sovereign debt default. No difference with observations in Table 4. *** p<0.01, ** p<0.05, * p<0.1.

Table 16: Resource-backed loans and sovereign debt default: using OLS regression

	[1]	[2]
VARIABLES	OLS standard	Driscoll-Kraay
Resource-backed loans	**1.5294*** **	**1.5294*** **
	(0.2271)	(0.2930)
Lag inflation	-0.0019	-0.0019
	(0.0036)	(0.0034)
Lag GDP per capita	-0.0117*	-0.0117
	(0.0067)	(0.0071)
Lag natural resources rents	-0.0143	-0.0143
	(0.0100)	(0.0089)
Lag fiscal balance	0.0232***	0.0232**
	(0.0076)	(0.0086)
Lag public debt	0.0102***	0.0102***
	(0.0014)	(0.0018)
Lag Domestic credit to private sector	0.0052	0.0052
	(0.0037)	(0.0034)
Lag Broad money	-0.0007	-0.0007
	(0.0031)	(0.0023)
Constant	6.5537***	3.1711***
	(0.6132)	(0.2462)
Observations	2,222	2,222
R-squared	0.7599	0.1479
Country/FE	Yes	Yes
Year/FE	Yes	Yes

Notes : This Table presents the effect of resource-backed loans on sovereign debt default obtained by OLS estimates. The treatment variable is resource-backed loan. The outcome variable is sovereign debt default. No difference with observations in Table 4. *** p<0.01, ** p<0.05, * p<0.1.

Table 17: Resource-backed loans and sovereign debt default: using GMM estimator

VARIABLES	System GMM
Lagged sovereign debt default	0.684***
	(0.012)
Resource-backed loans	0.661***
	(0.075)
Lag inflation	-0.013***
	(0.003)
Lag GDP per capita	-0.039***
	(0.007)
Lag natural resources rents	0.003
	(0.004)
Lag fiscal balance	0.024***
	(0.008)
Lag public debt	0.009***
	(0.001)
Lag Domestic credit to privates	-0.014***
	(0.002)
Lag Broad money	-0.004***
	(0.001)
Constant	1.256***
	(0.113)
Observations	2,222
Number of countries	119
Number of instruments	88
AR(1) test, p-value	0.000
AR(2) test, p-value	0.594
Hansen, p-value	0.797
Sargan, p-value	0.000

This table reports estimates of the impact of resource-backed loans on sovereign debt default using a two-step system-GMM and relying on internal instruments. The study period is 1990–2020. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 18: Resource-backed loans and sovereign debt default: Placebo test

VARIABLES	[1]	[2]	[3]	[4]	[5]
Placebo resource-backed loans: (random dates)	-0.2862	0.3542	0.0637	0.0876	-0.0036
	(0.5173)	(0.4292)	(0.3310)	(0.4410)	(0.2980)
Constant	5.1924***	5.4953***	5.8125***	4.5869***	6.7678***
	(0.1425)	(0.3314)	(0.2305)	(0.9209)	(0.8188)
Observations	2,222	2,222	2,222	2,222	2,222
R-squared	0.0008	0.2953	0.7837	0.3398	0.7982
Covariates	No	Yes	Yes	Yes	Yes
Country/FE	No	No	Yes	No	Yes
Year/FE	No	No	No	Yes	Yes

Notes : This Table presents the effect of resource-backed loans on sovereign debt default obtained by entropy balancing method. The treatment variable is resource-backed loan (random dates). The outcome variable is sovereign debt default. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 19: Resource-backed loans and sovereign debt default: Placebo test

VARIABLES	[1]	[2]	[3]	[4]	[5]	[6]
Random assignment	10%	20%	30%	40%	50%	60%
Placebo resource-backed loans	-0.0838	0.0064	0.0096	-0.0405	-0.0658	-0.0284
	(0.1625)	(0.1383)	(0.1204)	(0.1196)	(0.1160)	(0.1227)
Constant	6.8498***	6.7643***	6.7612***	6.7883***	6.8059***	6.7790***
	(0.8354)	(0.8208)	(0.8207)	(0.8271)	(0.8294)	(0.8207)
Observations	2,222	2,222	2,222	2,222	2,222	2,222
R-squared	0.7983	0.7982	0.7982	0.7983	0.7983	0.7982
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes : This Table presents the effect of resource-backed loans on sovereign debt default obtained by entropy balancing method. The treatment variable is resource-backed loan (random assignment). The outcome variable is sovereign debt default. *** p<0.01, ** p<0.05, * p<0.1.

Table 20: Resource-backed loans and sovereign debt default:Constraining the treatment

	[1]	[2]	[3]	[4]
VARIABLES	*[-2, 2]*	*[-3, 3]*	*[-4, 4]*	*[-5, 5]*
Resource-backed loans adoption	1.6064***			
	(0.2470)			
Resource-backed loans adoption		1.5930***		
		(0.2783)		
Resource-backed loans adoption			2.0989***	
			(0.2749)	
Resource-backed loans adoption				1.3299***
				(0.3580)
Constant	7.0758***	7.0702***	7.1698***	6.9834***
	(0.8006)	(0.8016)	(0.7971)	(0.7992)
Observations	2,222	2,222	2,222	2,222
R-squared	0.8057	0.8053	0.8076	0.8032
Covariates	Yes	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes	Yes

Notes : This Table presents the effect of resource-backed loans on sovereign debt default obtained by entropy balancing method. The treatment variable is resource-backed loan (constraining treatment period). The outcome variable is sovereign debt default. No difference with baseline results in Table 4. *** p<0.01, ** p<0.05, * p<0.1.

Table 21: Validity of transmission channels

	[1]	[2]	[3]
	Financial market access		Quality of institution
VARIABLES	Borrowing costs	Debt/GDP	Corruption
Resource-backed loans	0.8402**	0.1324**	0.0103*
	(0.3383)	(0.0672)	(0.0059)
Lag inflation	-0.0188**	-0.0008	0.0000
	(0.0076)	(0.0008)	(0.0001)
Lag gdp per capita	0.0623***	0.0014	0.0006**
	(0.0149)	(0.0031)	(0.0003)
Lag natural resources rents	-0.0167	0.0068**	0.0001
	(0.0145)	(0.0027)	(0.0003)
Lag fiscal balance	0.0349**	-0.0169***	0.0002
	(0.0163)	(0.0041)	(0.0004)
Lag public debt	-0.0398***	0.0078***	-0.0001***
	(0.0029)	(0.0010)	(0.0000)
Lag Domestic credit to privates	0.0281***	0.0038***	-0.0004*
	(0.0071)	(0.0014)	(0.0002)
Lag Broad money	-0.0055	-0.0003	0.0001
	(0.0039)	(0.0009)	(0.0001)
Constant	10.2168***	2.3862***	0.0825***
	(0.3083)	(0.1702)	(0.0138)
Observations	1,292	2,248	2,138
R-squared	0.9041	0.8359	0.9371
Country/FE	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes

Notes : This Table presents the channels transmission obtained by entropy balancing method. Borrowing costs are 5-year sovereign CDS spreads and debt viability is debt/exports. *** p<0.01, ** p<0.05, * p<0.1.

Table 22: Resource-backed loans and sovereign debt default: disaggregating RBLs

	[1]	[2]	[3]
VARIABLES			
Oil resource-backed loans	1.6701***		
	(0.2566)		
Minerals resource-backed loans		0.8558**	
		(0.4114)	
Other resource-backed loans			1.7411***
			(0.4548)
Constant	7.1790***	6.8076***	7.0584***
	(0.8232)	(0.8209)	(0.8120)
Observations	2,222	2,222	2,222
R-squared	0.8110	0.8000	0.8008
Covariates	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes

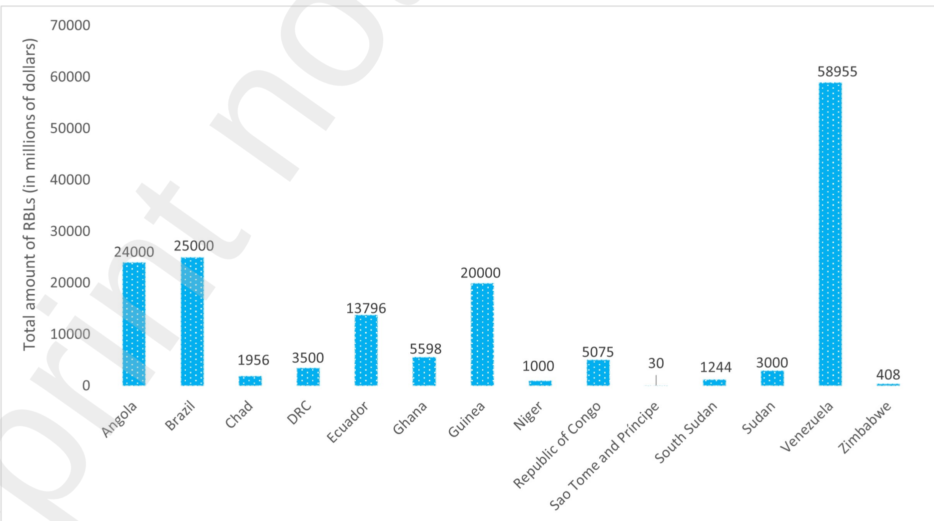
Notes : This Table presents the results of desegregated resource-backed loans obtained by entropy balancing method. Oil resource-backed loans are oil-backed, minerals resource-backed loans are mineral-backed (Copper, cobalt, bauxite, platinum, diamond, gold, etc.), and other resource-backed loans are cocoa-backed loans and tobacco. *** p<0.01, ** p<0.05, * p<0.1.

Figure 3: **Sovereign debt default and resource-backed loans.**

Cumulative RBL Debt (2004-2020) and Gross Government Debt					
Country	Debt/GDP	RBL/GDP	Sovereign debt default (USD)	RBL/Debt (% of Debt)	Risk of Debt Distress
Angola	54.266412	6.2281641	1491.6532	11.4770147324279	N/A
Brazil	71.190294	0.34984704	354.89435	0.491425193439993	N/A
Chad	36.192706	7.1505775	406.27896	19.7569573825179	High
Democratic Republic of the Congo	46.222353	8.8587912	930.93262	19.165599812714	Moderate
Ecuador	33.702235	2.7160167	1606.0276	8.05886226833324	N/A
Ghana	42.425647	3.4740363	458.26689	8.18852874536009	In debt distress
Guinea	55.202	192.2892	310.38512	348.337379080468	Moderate
Niger	28.503353	13.03781	140.64803	45.741320328173	Moderate
Republic of Congo	75.340529	13.675659	1856.5107	18.1517958282454	In debt distress
Sao Tome and Principe	120.03353	15.228426	93.562581	12.686810093813	In debt distress
South Sudan	52.6984	7.1639709	1064.7778	13.5942854052495	High
Sudan	111.33829	5.0471063	39712.982	4.53312719281031	In debt distress
Venezuela	94.39729	23.3	16848.2652228	24.682912	N/A
Zimbabwe	52.1805	1.437938	4361.5562	2.75569992621765	In debt distress

Source: Author's elaboration using data from Beers and Mavalwalla (2017) and Mihalyi et al. (2020). ***Notes*** : For the entire list of countries at risk of debt distress and eligible for debt relief, see "Debt Sustainability Analysis," World Bank and IMF via "Risk of debt distress".

Figure 4: **Total amount of resource-backed loans by country.**



Source: Author's elaboration using data from Mihalyi et al. (2020).

Table 23: Resource-backed loans and sovereign debt default : structural factors

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
VARIABLES								
Macroeconomic factors								
Resource-backed loans (RBL)	1.7481***	1.3027***	1.4982***	1.1464***	1.7752***	1.1040***	1.9325***	1.5118*
	(0.2996)	(0.2370)	(0.2290)	(0.2598)	(0.3317)	(0.2734)	(0.4621)	(0.8015)
RBL*government expenditure volatility	-0.1365*							
	(0.0730)							
Government expenditure volatility	0.0162***							
	(0.0059)							
RBL*inflation targeting (soft)		-1.3916**						
		(0.5900)						
Inflation targeting (soft)		-1.1174***						
		(0.3648)						
RBL*FDI			-0.0253*					
			(0.0152)					
FDI			0.0199***					
			(0.0062)					
RBL*remittance inflows				0.1545***				
				(0.0581)				
Remittance inflows				-0.0204*				
				(0.0105)				
RBL*resource depletion					-0.0384**			
					(0.0161)			
Resource depletion					0.0247**			
					(0.0101)			
Political and institutional factors								
Democracy						-0.6663*		
						(0.3670)		
RBL*democracy						-1.0471*		
						(0.6093)		
Rule of low							-0.6803**	
							(0.2870)	
RBL*rule of low							-0.7936*	
							(0.4115)	
Political stability								-0.9273
								(0.6991)
RBL*political stability								-0.5067
								(1.8566)
Observations	1,666	2,222	2,211	2,171	2,132	843	1,850	2,115
R-squared	0.7894	0.8070	0.8061	0.8093	0.8006	0.8439	0.8236	0.8148
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Undeclared constant included. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Note: The differences between columns [1], [3], [4], [5], [6], [7] and [8] in this Table and Table 4 are attributable to differences in observations on the interaction variables.

Table 24: Resource-backed loans and sovereign debt default : using IPW

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
ATT	1.6926***	2.0825***	2.1199***	1.9606***	1.9499***	2.1200***	1.3316***	1.8570***	1.6927***
	(0.2529)	(0.2473)	(0.3043)	(0.3235)	(0.2572)	(0.2350)	(0.2724)	(0.2979)	(0.2525)
Lag Inflation rate	-0.0036	-0.0006	-0.0003	-0.0008	-0.0024	-0.0003	-0.0049	0.0002	0.0002
	(0.0046)	(0.0043)	(0.0070)	(0.0072)	(0.0058)	(0.0061)	(0.0047)	(0.0055)	(0.0051)
Lag GDP per capita	-0.0341*	-0.0388**	-0.0550***	-0.0541**	-0.0403**	-0.0551***	-0.0410**	-0.0295	-0.0281
	(0.0186)	(0.0155)	(0.0205)	(0.0241)	(0.0183)	(0.0200)	(0.0204)	(0.0186)	(0.0176)
Lag Natural resources rents	0.0542***	0.0565***	0.0627***	0.0663***	0.0729***	0.0627***	0.0576***	0.0304***	0.0478***
	(0.0062)	(0.0081)	(0.0084)	(0.0086)	(0.0076)	(0.0085)	(0.0067)	(0.0076)	(0.0059)
Lag Fiscal balance	-0.1008***	-0.1090***	-0.0893***	-0.1022***	-0.1111***	-0.1043***	-0.1070***	-0.0865***	-0.1034***
	(0.0276)	(0.0312)	(0.0271)	(0.0288)	(0.0273)	(0.0303)	(0.0286)	(0.0272)	(0.0229)
Lag Public debt	-0.0033**	-0.0024	-0.0024	-0.0045	-0.0029	-0.0024	-0.0043**	0.0002	0.0014
	(0.0017)	(0.0019)	(0.0027)	(0.0036)	(0.0020)	(0.0029)	(0.0018)	(0.0016)	(0.0012)
Lag Domestic credit to private sector	-0.0193***	-0.0134***	-0.0187***	-0.0179***	-0.0314***	-0.0187***	-0.0201***	-0.0177***	-0.0265***
	(0.0058)	(0.0050)	(0.0064)	(0.0059)	(0.0064)	(0.0052)	(0.0056)	(0.0060)	(0.0053)
Lag Broad money growth	0.0140***	0.0095	0.0073	0.0080	0.0081	0.0074	0.0158***	0.0124**	0.0136**
	(0.0053)	(0.0062)	(0.0064)	(0.0094)	(0.0067)	(0.0081)	(0.0050)	(0.0051)	(0.0059)
Lag Financial openness		-0.2239**							
		(0.0879)							
Lag Government expenditure			0.0149						
			(0.0111)						
Lag Interest on debt				0.0936*					
				(0.0565)					
Lag Financial development					3.2530*				
					(1.8749)				
Lag Government revenue						0.0149			
						(0.0093)			
Lag Terms of Trade Index							-0.0044***		
							(0.0017)		
Lag Terms of Trade Index (world bank)								-0.0171***	
								(0.0018)	
Lag Crude oil brent (log)									-1.5792***
									(0.1586)
Constant	-3.1140***	-3.4220***	-3.5333***	-3.3596***	-3.6161***	-3.5336***	-2.5561***	-5.2933***	-9.7261***
	(0.2186)	(0.2549)	(0.4335)	(0.2954)	(0.3968)	(0.4286)	(0.3028)	(0.3983)	(0.7593)
Observations	2,222	2,090	1,732	1,731	2,171	1,732	2,171	2,105	2,222

Standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Table 25: Descriptive statistics of the variables from the baseline model

Variable	Obs	Mean	Std. Dev.	Min	Max
Sovereign debt default total	4075	3.71	3.163	0	12.652
RBLs	4153	0.04	0.195	0	1
RBL Oil	4153	0.008	0.09	0	1
RBL Mineral	4153	0.001	0.038	0	1
RBL Other	4153	0	0.022	0	1
Inflation rate	3438	36.443	466.883	-16.117	23773.131
GDP per capita	3817	2.085	6.734	-64.426	140.48
Natural resources rents	3844	7.732	11.032	0	87.577
Fiscal balance	3437	-2.421	5.796	-59.742	125.135
Public debt	3009	59.088	48.148	0.488	600.13
Credit to private sector	3055	32.264	30.276	0	255.31
Broad money growth	3430	34.064	253.474	-99.864	7677.834

Table 26: List of resource-backed loans

Country	Loan year	Loans in millions of USD	Resources used	loans country	loans entity	Sectors targeted for investment
Angola	2004	2000	Oil	China	Eximbank	Infrastructure
Angola	2007	500	Oil	China	Eximbank	Infrastructure
Angola	2007	2000	Oil	China	Eximbank	Infrastructure
Angola	2009	2000	Oil	China	Eximbank	Infrastructure
Angola	2010	2500	Oil	China	ICBC	housing
Angola	2015	15000	Oil	China	CDB	Infrastructure
Brazil	2009	1000	Oil	China	CDB	Oil
Brazil	2015	3500	Oil	China	CDB	Oil
Brazil	2015	1500	Oil	China	CDB	Oil
Brazil	2016	5000	Oil	China	CDB	Budget support and debt rollover
Brazil	2017	5000	Oil	China	CDB	Oil
Chad	2013	600	Oil	International	Glencore	Budget support and debt refinancing
Chad	2014	1356	Oil	International	Glencore	Oil
RDC	2008	3000	Copper & Cobalt	China	Eximbank	Infrastructure
RDC	2011	500	Copper	Corée	Korea Exim	Infrastructure
Ecuador	2010	1000	Oil	China	CDB	Infrastructure
Ecuador	2011	2000	Oil	China	CDB	Infrastructure
Ecuador	2012	2000	Oil	China	CDB	Budget support and debt rollover
Ecuador	2015	5296	Oil	China	Eximbank	Infrastructure
Ecuador	2015	1500	Oil	China	CDB	Infrastructure
Ecuador	2016	1500	Oil	China	CDB	Infrastructure
Ecuador	2016	500	Oil	China	CDB	Budget support and debt rollover
Ghana	2011	1500	Oil	China	CDB	Infrastructure
Ghana	2011	1500*	Oil	China	CDB	Infrastructure
Ghana	2018	2000	Bauxite	China	Sinohydro	Infrastructure
Guinea	2017	20000	Bauxite	China	CCC	Infrastructure
Niger	2013	1000*	Oil	China	Eximbank	Oil
Republic of Congo	2006	1600	Oil	China	Eximbank	Infrastructure
Republic of Congo	2011	625	Oil	International	Gunvor	Oil
Republic of Congo	2012	1000	Oil	China	Eximbank	Infrastructure
Republic of Congo	2015	1000	Oil	International	Trafigura	Unknown
Republic of Congo	2015	850	Oil	International	Glencore	Unknown
Sao Tome and Principe	2010	30	Oil	Nigeria	Gouvernement	Oil
South Sudan	2015	75	Oil	International	CNPC	Unknown
South Sudan	2015	1000	Oil	China	Eximbank	Budget support
South Sudan	2016	169	Oil	China	Eximbank	Road
Sudan	2007	3000	Oil	China	Eximbank	Infrastructure
Venezuela	2006	6500	Oil	Russia	Rosneft	Infrastructure
Venezuela	2007	4000	Oil	China	CDB	Infrastructure
Venezuela	2009	4000	Oil	China	CDB	Infrastructure
Venezuela	2010	20255	Oil	China	CDB	Infrastructure
Venezuela	2011	4000	Oil	China	CDB	Infrastructure & Oil
Venezuela	2013	4000	Oil	China	CDB	Oil
Venezuela	2013	5000	Oil	China	CDB	Infrastructure
Venezuela	2014	4000	Oil	China	Eximbank	Infrastructure
Venezuela	2015	5000	Oil	China	CDB	Infrastructure
Venezuela	2016	2200	Oil	China	CDB	Oil
Zimbabwe	2004	110	Tobacco	China	CATIC	Power
Zimbabwe	2006	200	platinum	China	Eximbank	Agricultural
Zimbabwe	2011	98	Diamond	China	Eximbank	Education

Source: Author's construction based on information from Mihalyi et al. (2022). Note: RBLs marked with * were subsequently cancelled without disbursement.

Table 27: Robustness: Additional control variables

VARIABLES	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]
RBLs	1.5150***	2.6134***	2.0879***	2.4942***	1.3333***	1.5420***	1.3400***	1.6242**	1.5148***	1.8953***	1.4834***	1.4897***	1.0510***
	(0.2332)	(0.5762)	(0.6005)	(0.5638)	(0.2188)	(0.2311)	(0.2127)	(0.6972)	(0.2223)	(0.2557)	(0.2537)	(0.2717)	(0.2241)
Debt service total	0.0478**												
	(0.0214)												
Inflation crisis		0.5502											
		(0.6332)											
Banking crisis			0.3488										
			(1.2044)										
Currency crises				1.0583**									
				(0.4867)									
Fiscal rules					-1.1891***								
					(0.2208)								
Monetary policy volatility						0.0093**							
						(0.0045)							
GDP per capita shocks							0.0151						
							(0.0124)						
Credit rating								-0.0038					
								(0.1232)					
Inflation shocks									0.0029*				
									(0.0017)				
Fiscal discipline volatility										-0.0248			
										(0.0327)			
Financial openness											-0.0905		
											(0.1295)		
Exchange rate												-0.0245	
												(0.0516)	
EITI countries													-1.6596***
													(0.3953)
Constant	7.9499***	2.6227*	4.4376**	3.7217**	6.6434***	8.6780***	8.1280***	1.6976	8.1494***	1.3898	0.8654	7.1176***	8.6772***
	(0.7462)	(1.4989)	(2.1734)	(1.6987)	(0.7599)	(0.9374)	(0.9527)	(1.6101)	(0.9531)	(0.9031)	(0.8787)	(0.9090)	(0.4551)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,988	607	530	610	2,250	2,149	2,189	1,260	2,146	1,571	2,118	1,810	1,859
R-squared	0.7983	0.7281	0.6992	0.7361	0.8041	0.8037	0.8085	0.7349	0.8178	0.8171	0.7797	0.8161	0.8313

Robust standard errors in parentheses *** $p<0.01$, ** $p<0.05$, * $p<0.1$

Table 28: List of resource-backed loans (RBL) and Non-RBL countries

RBL	**non-RBL**	**non-RBL**	**non-RBL**	**non-RBL**	**non-RBL**
Angola	Bhutan	Fiji	Malawi	Rwanda	Vanuatu
Brazil	Bolivia	Gabon	Maldives	Samoa	Vietnam
Chad	Bosnia & Herzegovina	Gambia	Mali	Senegal	Yemen
Congo, Dem Rep	Botswana	Georgia	Mauritania	Serbia	Zambia
Congo, Rep	Bulgaria	Greece	Mauritius	Seychelles	
Ecuador	Burkina Faso	Grenada	Mexico	Sierra Leone	
Ghana	Burundi	Guatemala	Moldova	Slovenia	
Guinea	Cambodia	Guinea-Bissau	Mongolia	Solomon Islands	
Niger	Cameroon	Guyana	Montenegro	Somalia	
Sao Tome and Principe	Central African Republic	Haiti	Morocco	South Africa	
South Sudan	Chile	Honduras	Mozambique	Sri Lanka	
Sudan	Colombia	India	Myanmar	St. Lucia	
Venezuela	Comoros	Indonesia	Nauru	Suriname	
Zimbabwe	Costa Rica	Iran	Nepal	Syria	
non-RBL	**non-RBL**	**non-RBL**	**non-RBL**	**non-RBL**	
Afghanistan	Cote d'Ivoire	Iraq	Nicaragua	Tajikistan	
Albania	Croatia	Ireland	Nigeria	Tanzania	
Algeria	Cuba	Jamaica	Pakistan	Thailand	
Antigua and Barbuda	Cyprus	Jordan	Panama	Togo	
Argentina	Djibouti	Kazakhstan	Papua New Guinea	Tonga	
Armenia	Dominica	Kenya	Paraguay	Tunisia	
Azerbaijan	Dominican Republic	Kyrgyz Republic	Peru	Turkey	
Bangladesh	Egypt	Lebanon	Philippines	Turkmenistan	
Barbados	El Salvador	Lesotho	Poland	Uganda	
Belarus	Equatorial Guinea	Liberia	Portugal	Ukraine	
Belize	Eritrea	Libya	Puerto Rico	Uruguay	
Benin	Ethiopia	Madagascar	Romania	Uzbekistan	

Table 29: Sources, and definitions of variables

Variables	Definition	Sources
1. Dependent variable		
Sovereign debt default	The face value of sovereign government debt in default.	Beers and Mavalwalla (2017)
Sovereign debt default dummy	Takes value 1 if country i at period t has defaulted on its sovereign debt	Balima et al. (2017); Jorra (2012)
2.Treatment variable		
Resource-backed loans (RBLs)	Takes value 1 if country i at period t has signed a resource-backed loans	Mihalyi et al. (2020)
Oil resource-backed loans	Takes value 1 if country i at period t has signed a loans on oil	Mihalyi et al. (2020)
Minerals resource-backed loans	Takes value 1 if country i at period t has signed a loans on minerals	Mihalyi et al. (2020)
Other resource-backed loans	Takes value 1 if country i at period t has signed a loans on tobacco and cocoa	Mihalyi et al. (2020)
3. Baseline model control variables or covariates		
Inflation rate	Inflation, average consumer prices (Percent change)	WDI
Natural resources rents	Natural resources rents to % GDP	WDI
GDP per capita	GDP per capita is gross domestic product (constant 2010 U.S. dollars)	WDI
Fiscal balance	Fiscal balance (% of GDP)	Kose et al. (2017)
Public debt	Public debt (% of GDP)	Ali Abbas et al. (2011)
Credit	Domestic credit to private sector (% of GDP)	WDI
Broad money	Broad money growth (% of GDP)	WDI
4. Additional control variables		
Financial development	Financial development index	IMF Financial Development database
Total expenditure (% of GDP)	Total expenditures include total expenditures and net acquisition of non-financial assets	World Economic Outlook (WEO)
Interest on debt	Interest paid on public debt, percent of GDP	IMF
Government revenue	Government revenue, percent of GDP (% of GDP)	IMF
Oil prices	Crude oil brent	World Bank Commodity Price Data
Terms of trade index IMf	Commodity prices indices, including export, import, and terms-of-trade indices	IMF-WEO
Terms of trade index WDI	It captures the probability of default of a country's commercial banking system.	WDI
Corruption	Corruption measures perceptions of corruption, conventionally defined as the exercise of public power for private gain	Teorell et al. (2020)
Government expenditure volatility	Standard deviation of the government expenditure estimated over 5-year moving window	Authors' calculation based on WDI data
Inflation targeting (soft)	Dummy	Rose (2007); Roger (2009); Tapsoba et al. (2012); Jahan (2012); Ciżkowicz-Pękała et al. (2019)
FDI	Foreign direct investment, net inflows (% of GDP)	WDI
Remittances flows	Personal remittances comprise personal transfers and compensation of employees	WDI
Resource depletion	Natural resource depletion is the sum of net forest depletion, energy depletion, and mineral depletion	WDI
Rule of law	Rule of Law includes several indicators which measure the extent to which agents have confidence in and abide by the rules of society	Teorell et al. (2020)
Democracy	Index ranging from -10 to 10	Policy 2
Political stability	Index ranging from -2 to 2	WGI, Kaufmann et al. (2011)
Debt service total	Total service debt (% of GDP)	WDI
Inflation crisis	Dummy variable equal to one if the annual inflation rate is 20% or higher	Reinhart and Rogoff (2009)
Fiscal policy volatility (revenue)	Standard deviation of the government revenue estimated over 5-year moving window	Authors' calculation based on WDI data
Monetary policy volatility	Standard deviation of the broad money estimated over 5-year moving window	Authors' calculation based on WDI data
Inflation shocks	Standard deviation of the inflation rate estimated over 5-year moving window	Authors' calculation based on WDI data
Corruption control	Index ranging from 0 to 100	WGI, Kaufmann et al. (2011)
Banking crisis	Dummy variable equal to one for a banking crisis.	Reinhart and Rogoff (2009)
Currency crises	Dummy variable equal to one if at least one financial crisis occurs	Reinhart and Rogoff (2009)
Fiscal rules	Dummy	Reinhart and Rogoff (2009)
GDP per capita shochs	Standard deviation of the GDP per capita estimated over 5-year moving window	IMF dataset, Lledó et al. (2017)
Credit rating	Index ranging from 1 to 21	Kose et al. (2017))
Fiscal discipline volatility	Standard deviation of the Cyclically-adjusted balance, % of potential GDP estimated over 5-year moving window	Authors' calculation based on WDI data
Financial openness	Capital Account Openness index	Chinn and Ito (2006)
Exchange rate	Exchange rate, national currency/USD (market+estimated)	Penn World Table 10.0
EITI countries	Takes the value of 1 starting from when a country becomes an EITI member and 0 for the years in which the government is not an EITI member	World Bank Group.2016 (2016); Kinda and Thiombiano (2024)